2-D Precise Controllable HRWS-SAR Smart Jamming: Through Segmented-Phase Modulation

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***Abstract*— Azimuth multichannel synthetic aperture radar (SAR) possesses high-resolution and wide-swath (HRWS) imaging capabilities, posing a significant threat to the survivability of critical targets during wartime. However, due to the constraints of multichannel reconstruction processing, existing jamming technology will produce a periodic jamming effect in the azimuth direction when against HRWS-SAR. This unique periodic repetition related to the number of channels will make the jamming easy to detect and eliminate. Meanwhile, the amplitude of the jamming signal suffers from the modulation of sine functions, which causes a waste of jamming energy. Against the above problems, this article introduces a presegmentation idea, effectively circumventing the aforementioned problems. Meanwhile, this article proposes a 2-D precise controllable smart jamming method designed for HRWS-SAR. The proposed method can generate two jamming effects, barrage and deceptive, at designated locations in the targeted area. First, 2-D expand factors are defined, respectively, to achieve precise control of the barrage jamming range. Then, by controlling multiple jamming parameters, a highly controllable deceptive jamming effect can be achieved. Additionally, a parameterized jamming model is established, which can achieve 2-D precise control of the jamming position through the cooperation of the signal and spatial domain. Notably, this study focuses not only on the impact of different jamming parameters on imaging results but also on evaluating jamming effectiveness under parameter estimation errors to guide optimal jamming strategies. Finally, extensive numerical simulation experiments and comparisons with other methods are conducted to verify the superiority of the proposed method.**

***Index Terms*— Barrage jamming, deceptive jamming, high-resolution and wide-swath (HRWS)-synthetic aperture radar (SAR), segmented phase modulation.**

I. Introduction

A. Background

SYNTHETIC aperture radar (SAR), as an active microwave sensor, has imaging advantages including day and night operability, weather independence, and high resolution [1], [2]. The excellent imaging capabilities of SAR have made it shine in military applications such as topographic surveying, intelligence collection, battlefield surveillance, and so on [3], [4]. Meanwhile, increasingly complex detection needs have promoted the birth of SAR systems with different working modes, such as Scan-SAR [5], Bistatic SAR [6], and high-resolution and wide-swath SAR (HRWS-SAR) [7]. HRWS-SAR attracted much more attention because it overcomes the restriction of the smallest antenna area. Specifically, HRWS-SAR achieves a wide swath in the range direction through a lower pulse repetition frequency (PRF) and suppresses Doppler blur by arranging multiple receiving channels [8], [9]. Many spaceborne and airborne SARs have been equipped with this working mode, such as Radarsat-2 [10], TerraSAR-X [11], Gaofen-3 [12], and so on. However, the considerable application of HRWS-SAR systems and the further improvement of imaging capabilities have made electronic countermeasures (ECMs) face new uncertainties and difficulties [13], [14], [15]. Therefore, to protect the regions of interest (ROIs) from observation and detection, it is urgent to propose corresponding jamming methods against the imaging characteristics of HRWS-SAR.

Received 9 September 2024; revised 20 November 2024; accepted 9 February 2025. Date of publication 18 February 2025; date of current version 17 June 2025. This work was supported in part by Shenzhen Science and Technology Program under Grant KQTD20190929172704911, in part by the Science and Technology Planning Project of Key Laboratory of Advanced IntelliSense Technology, in part by Guangdong Science and Technology Department under Grant 2023B1212060024, and in part by the National Natural Science Foundation of China under Grant 61971429 and Grant 61871385. *(Corresponding author: Wei Wang.)*

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Digital Object Identifier 10.1109/TGRS.2025.3543006

B. Previous Work

Researchers have conducted extensive studies over the years against SAR jamming technology to discover effective and robust jamming methods [16], [17], [18]. Generally, these methods can be divided into two types: barrage [19] and deceptive [20].

1) Barrage Jamming: To cover specific targets in the image, barrage jammers typically increase the noise floor of the radar receiver by transmitting high-powered noise signals. However, a higher transmit power and cost budget are essential in actual ECM [21]. In recent years, to reduce the transmit power of jammers, a series of coherent jamming methods have been proposed [22]. In [23], the intercepted radar signals were retransmitted with random time delay. The barrage jamming method of noise convolution type produced the jamming signal by convolving the noise signal with the intercepted SAR signal [24].

The aforementioned techniques can obtain a certain processing gain in pulse compression (PC), which can be considered partially coherent jamming. Frequency shift jamming, including stepped frequency shift [25], random frequency shift [26], and segmentation random frequency shifting jamming [27],

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further improved the processing gain of jamming signals. However, due to the presence of additional frequency shift terms, the jamming energy becomes excessively dispersed. Subsequently, the concept of the multiple-phase-sectionalized modulation (MPSM) was introduced in [28] and [29]. The MPSM was implemented by dividing the intercepted linear frequency-modulated (LFM) signal into several segments in the time domain and adding random phases to each segment. Although the MPSM could produce noise-like jamming effects in the mainlobe range, the sidelobes will rise significantly. Against this insufficiency, [30] proposed an MPSM based on cosine amplitude weighting (M-CAW), which could suppress sidelobes and further improve the utilization of jamming energy. Regrettably, the jamming range of M-CAW was fixed. Based on this, through the controllable time–frequency relationship of the nonlinear FM signal [31], [32], Li et al. [32] achieved precise control of the jamming range in the down-range direction. However, they did not discuss the azimuth direction. Subsequently, Liu et al. [33] realized the expansion of the azimuth jamming range based on the phase error theory. Wang et al. [34] realized the 2-D control of the jamming range based on the quadratic phase error (QPE), and it was further applied in [35].

The above works have conducted a lot of discussion and research on SAR barrage jamming. However, most methods are limited to the range direction. With the continuous upgrading of ECM, 2-D joint precise controllable jamming in the range and azimuth directions received more attention.

2) Deceptive Jamming: Deceptive jamming implants an elaborately designed false scene or target into the SAR image by sending a false target signal with similar characteristics to the real target signal, thereby achieving the jamming effect of "hard-to-distinguish real from imitation" [36]. However, the high requirement of computational complexity and parameter reconnaissance accuracy is an unavoidable problem.

To alleviate the above problems, inverse imaging-type deceptive jamming methods, including the inverse range–Doppler (RD) [37], inverse Omega-K [38], and inverse chirp-scaling algorithm [39], generated deceptive jamming signals by applying the inverse process of imaging to a preset electromagnetic simulation model. However, this type of method required precise calculation of the flying position of the radar and transmission of false target jamming signals to the corresponding pulse moment, which was usually difficult to achieve in a high-speed and changeable war environment. Intermittent sampling retransmission jamming (ISRJ) achieves deceptive jamming effects by intermittently intercepting and retransmitting radar signals. The essence of false targets generated by ISRJ was due to periodic missing data in the time domain, which could be easily eliminated through data recovery technology [40]. Chen et al. [41] proposed a cosine-phase-modulation jamming method. However, the amplitude of the false target gradually decreases from the inside to the edge, making it easy to identify and eliminate. Frequency-diversity array (FDA) has smaller frequency increments between adjacent antenna elements, resulting in a transmit beam pattern that is dependent on angle and range. Based on the characteristics of the FDA, Huang et al. [42], [43] proposed a space–time–frequency-coordinated deceptive jamming method and conducted a detailed analysis. Unfortunately, false targets only exist in the down-range direction. Recently, a periodic phase modulation jamming (PPMJ) method has attracted attention, which can generate deceptive jamming by performing periodic phase modulation against intercepted SAR signal [44].

The problem of low controllability and strong distribution regularity of false targets has still not been solved well. In addition, a 2-D flexible and controllable deceptive jamming effect is difficult to achieve, which limits the flexibility of jamming implementation.

The above-mentioned research, involving barrage techniques and deceptive measures, has been confirmed to exert a certain level of jamming effects against SAR. It is worth noting that the majority of these methods are proposed in the context of single-channel SAR systems. In the realm of multichannel systems, Dong [45] delves into the imaging attributes affected by Doppler-modulation jamming in HRWS-SAR. However, the distance difference between channels was not considered. In [46], the imaging properties of rebound jamming against HRWS-SAR were analyzed. In [47], the noise on HRWS-SAR imaging results was analyzed, where noise is distributed over the SAR image after imaging. Wang et al. [48] proposed a multitarget barrage jamming technology for HRWS-SAR but did not consider the impact of multichannel reconstruction processing. Ji et al. [49] comprehensively analyzed the imaging characteristics of jamming signals with Doppler-modulation jamming passing through HRWS-SAR for the first time. The analysis results indicate that after multichannel reconstruction in the azimuth direction, the jamming signal will exhibit periodic spectrum aliasing, with its amplitude modulated by a sine function [50]. This repetitive spectrum pattern would lead to a discernible distribution of jamming across the HRWS-SAR image, heightening the risk of identification. Moreover, the modulation of the sine function leads to additional wastage of jamming energy.

Overall, these analytical studies provide valuable insights that enhance the theoretical basis for future research on ECM against HRWS-SAR [51]. However, it is undeniable that there is currently a lack of effective jamming methods specifically designed against HRWS-SAR, which may seem surprising.

C. Main Works and Contributions

Against the aforementioned challenges, this article presents a 2-D precise controllable HRWS-SAR smart jamming method through segmented-phase modulation (PCHS-SP). By introducing the idea of presegmentation, PCHS-SP effectively circumvents the periodic repetition of the jamming signal spectrum problem and mitigates the modulation-induced amplitude loss. Simulation experiments confirm the effectiveness of the proposed method.

The main contributions of this article are summarized as follows.

1) The imaging characteristics of jamming signals under high-resolution SAR are studied. This article proposes a 2-D precise controllable HRWS-SAR smart jamming

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method, that is, PCHS-SP. PCHS-SP avoids the insufficient encountered by traditional Doppler-modulation jamming when countering HRWS-SAR and is implemented simply and can produce a barrage or deceptive jamming effect.

2) Based on MPSM jamming, we proposed range and azimuth expand factors and established a mapping relationship between the 2-D modulation factor and the jamming range, which can achieve a 2-D precise controllable jamming range. In addition, the application and discussion of MPSM in the azimuth direction are given for the first time.
3) A parameterized jamming geometry model is established to achieve 2-D precise control of the jamming position. The precise control of the jamming position in terms of azimuth and range is achieved by combining delayed retransmission and frequency shift modulation.
4) This article not only thoroughly analyzes the influence of different jamming parameters for the jamming effect. Meanwhile, we pay more attention to the jamming effect with the presence of parameter measurement errors to determine appropriate ECM decisions. In Section V, the imaging characteristics of jamming methods under the HRWS-SAR system, such as noise convolution and phase error modulation (PEM), are analyzed for the first time. Meanwhile, these methods are used as comparative experiments to comprehensively illustrate the effectiveness and superiority of the proposed method.

D. Organization

The remainder of this article is organized as follows. In Section II, a parameterized jamming geometry model is introduced, while the imaging characteristics of the Doppler-modulation jamming signal are analyzed for HRWS-SAR. In Section III, a 2-D precise controllable HRWS-SAR smart jamming method is introduced in detail. Section IV analyzes the impact of different parameters on the jamming effect and the jamming characteristics under the presence of reconnaissance errors. In Section V, the simulation results of the experiment are shown, and the effectiveness of the proposed method is verified. Finally, conclusions are drawn in Section VI.

II. Parameterized Model and Basic Theories

A. Parameterized Geometric Model

Fig. 1 shows the working geometric model of the SAR and the jammer. According to the right-handed coordinate system criterion, the XOY rectangular coordinate system is established, with O serving as the origin. Assume that the opponent's radar flight altitude is with h_0 and along the X-axis with the speed v_a. The X- and Y-axes represent the azimuth and ground-range directions, respectively.

The working mechanism of jammer implementation is based on "interception–modulation–retransmit." The radar system transmits signals to ROIs at time t_1. The jammer is deployed at point $P(x_p, y_p, z_p)$ to conduct reconnaissance and interception of the transmitted radar signals and obtain some parameter

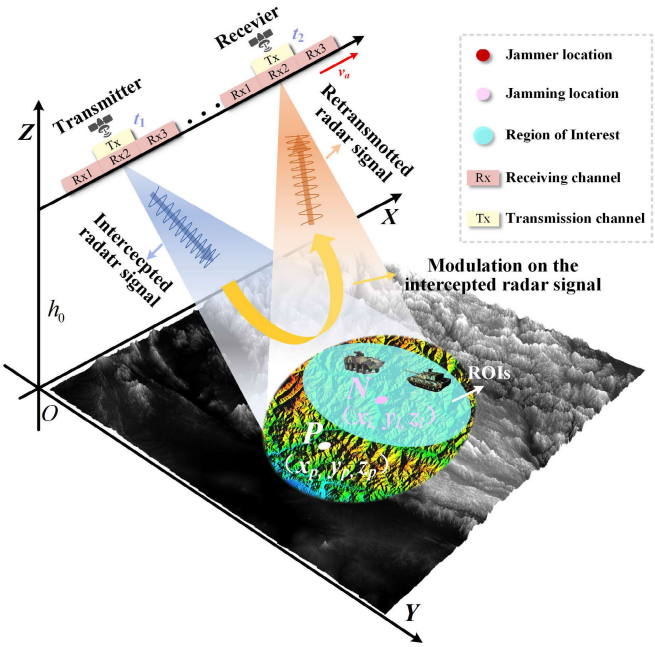


Fig. 1. Parameterized jamming geometry model.

information of the interception signal. It is particularly worth noting that different from the traditional geometric model, the jammer proposed in this article is not deployed in the center area of the ROIs, but is deployed at a distance from the front of the ROIs. This placement has the following two advantages.

1) The work delays of the jammer can be compensated by strategically positioning it in specific domain locations.
2) Based on the prior information with the position difference between the jammer and the preset jamming position $N(x_i, y_i, z_i)$, precise control of the azimuth jamming position in the ROIs can be easily achieved through delayed retransmitting without the need for complex signal modulation.

Then, the jammer modulates the intercepted SAR to generate a jamming signal with a barrage or deceptive jamming effect according to the selection of the modulation item. The generated jamming signal is retransmitted into the SAR system through the jammer at time t_2. Additionally, given its countermeasures to HRWS-SAR, each channel in the azimuth direction will receive jamming signals. Finally, after the jamming signals received by different channels are processed by HRWS-SAR imaging, the jamming effects are generated at the designated locations in the image ROIs.

B. Imaging Theories of Doppler-Modulation Jamming Against HRWS-SAR

In this section, the imaging characteristics of Doppler-modulation jamming against HRWS-SAR are analyzed. Taking the three channels as an example, Fig. 2 shows the working principle of the HRWS-SAR system. In this configuration, the central channel serves as the transmitting aperture, while all channels receive ground echoes. The right half of Fig. 2 shows the principle of

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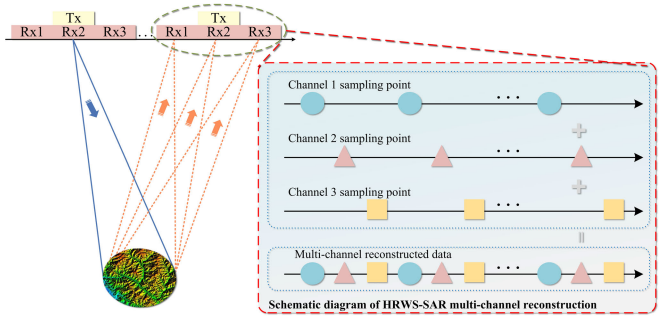


Fig. 2. Imaging schematic of HRWS-SAR.

multichannel reconstruction. The sampling points of each channel are arranged in a unique way to achieve the purpose of increasing PRF [52].

For HRWS-SAR, the range echo signal from each channel is consistent with that in the Stripmap mode. Consequently, for the sake of simplicity, this section only focuses on the azimuth echo signal of the HRWS-SAR system. Using a point target as an example, the jamming signal received by the mth channel can be represented as

$$S_m(\eta) = \omega(\eta)\exp\left[-j\frac{2\pi f_0 R_{Jm}(\eta)}{c}\right]p(\eta) \tag{1}$$

where $\omega(\eta)$ represents the azimuth envelope curve, f_0 represents the carrier frequency, and c is the speed of electromagnetic waves in free space. $R_{Jm}(\eta)$ denotes the instantaneous slant-range between the mth channel of the HRWS-SAR sensor and the jammer. $p(\eta)$ is the Doppler-modulation jamming term. The Doppler-modulation jamming here can be a variety of single-frequency or multifrequency modulation jamming such as frequency shift modulation and noise multiplication modulation jamming (NMMJ).

From the HRWS-SAR imaging mechanism, it can be known that $p(\eta)$ has the same expression in each channel, which can be regarded as a constant [49], [50]. After the modulation phase $p(\eta)$ of each receiving channel is discretized, which can be expressed as $\mathbb{X}(l), l = 0, 1, \ldots, L-1$. L represents the total number of sampling points in the azimuth direction, and the corresponding spectrum can be expressed as $\mathbb{X}(f_a)$. Assuming the Doppler-modulation jamming term after multichannel reconstruction processing, as shown in Fig. 2, is denoted as $\mathbb{Y}(l)$. The relationship between $\mathbb{X}(l)$ and $\mathbb{Y}(l)$ can be expressed as

$$\begin{aligned}\mathbb{Y}(l) = &\underbrace{\mathbb{X}(0), \mathbb{X}(0), \ldots, \mathbb{X}(0)}_{M}, \underbrace{\mathbb{X}(1), \mathbb{X}(1), \ldots, \mathbb{X}(1)}_{M},\\ &\ldots, \underbrace{\mathbb{X}(L-1), \mathbb{X}(L-1), \ldots, \mathbb{X}(L-1)}_{M}\end{aligned} \tag{2}$$

where M is the total number of receiving channels. The spectrum of $\mathbb{Y}(l)$ is usually complex. To simplify, the spectrum of (3) can be first solved

$$\begin{aligned}&\tilde{Y}(l)\\ &= \mathbb{X}(0), \underbrace{0, \ldots, 0}_{M-1}, \mathbb{X}(1), \underbrace{0, \ldots, 0}_{M-1}, \ldots, \mathbb{X}(L-1), \underbrace{0, \ldots, 0}_{M-1}.\end{aligned} \tag{3}$$

Periodic zero padding will cause periodic repetition of the spectrum. Hence, $\tilde{Y}(f_a)$ can be expressed as

$$\begin{aligned}\tilde{Y}(f_a) = &\mathbb{X}(f_a) \otimes \delta(f_a) + \mathbb{X}(f_a) \otimes \delta\left(f_a - \mathbf{F}_p\right) + \cdots\\ &+ \mathbb{X}(f_a) \otimes \delta\left[f_a - (M-1)\mathbf{F}_p\right]\end{aligned} \tag{4}$$

where $\delta(f_a)$ represents the impulse function. $\mathbf{F}_p$ is the actual PRF of the HRWS-SAR system. $\otimes$ denotes convolution operation. Comparing (2) and (3), $\mathbb{Y}(l)$ can be expressed by $\tilde{Y}(l)$ as

$$\mathbb{Y}(l) = \tilde{Y}(l) + \tilde{Y}(l-1) + \cdots + \tilde{Y}[l-(M-1)]. \tag{5}$$

Therefore, the spectrum of $\mathbb{Y}(l)$ can be derived as

$$\mathbb{Y}(f_a) = \tilde{Y}(f_a) \cdot \mathrm{g}(f_a) \tag{6}$$

where $\mathrm{g}(f_a)$ can be denoted as

$$\begin{aligned}&\mathrm{g}(f_a)\\ &= 2\exp\left[-2j\pi f_a \frac{(M-1)}{2M \cdot \mathbf{F}_p}\right]\left[\cos\left(2\pi f_a \frac{(M-1)}{2M \cdot \mathbf{F}_p}\right)\right.\\ &\left.+ \cos\left(2\pi f_a \frac{(M-3)}{2M \cdot \mathbf{F}_p}\right) + \cdots + \cos\left(2\pi f_a \frac{1}{2M \cdot \mathbf{F}_p}\right)\right].\end{aligned} \tag{7}$$

The above equation is the case when M is an even number. If M is the odd number, $\mathrm{g}(f_a)$ can be expressed as

$$\begin{aligned}\mathrm{g}(f_a) = &2\exp\left[-j2\pi f_a \frac{(M-1)}{2M \cdot \mathbf{F}_p}\right]\left[\cos\left(2\pi f_a \frac{(M-1)}{2M \cdot \mathbf{F}_p}\right)\right.\\ &\left.+ \cos\left(2\pi f_a \frac{(M-3)}{2M \cdot \mathbf{F}_p}\right) + \cdots + 1\right].\end{aligned} \tag{8}$$

In conclusion, after multichannel reconstruction, the spectrum of the Doppler-modulation jamming term will appear as periodic repetitions in the frequency domain and repeat $M-1$ times at $\mathbf{F}_p$ intervals. To further illustrate the imaging characteristics of Doppler-modulation jamming, we conducted a detailed theoretical analysis using NMMJ as an example. The spectrum of NMMJ can be modeled as a linear combination of the spectra of multiple single-frequency signals, which can be expressed as

$$\mathbf{N}(\eta) = \sum_{f_a^i=-B_n^a}^{B_n^a} \exp\left(j2\pi f_a^i \eta\right) \tag{9}$$

where B_n^a represents the number of frequencies point of noise. The spectrum of $\mathbf{N}(\eta)$ can be expressed as

$$\mathbf{N}(f_a) = \sum_{f_a^i=-B_n^a}^{B_n^a} \delta\left(f_a - f_a^i\right). \tag{10}$$

If $f_a^i > \mathbf{F}_p$, the spectrum of the Doppler-modulation jamming signal received by each receiving antenna will be folded. Therefore, the actual modulation frequency is

$$\ddot{f}_a^i = f_a^i - \lfloor \frac{f_a^i}{\mathbf{F}_p} - \frac{1}{2} \rfloor \mathbf{F}_p. \tag{11}$$

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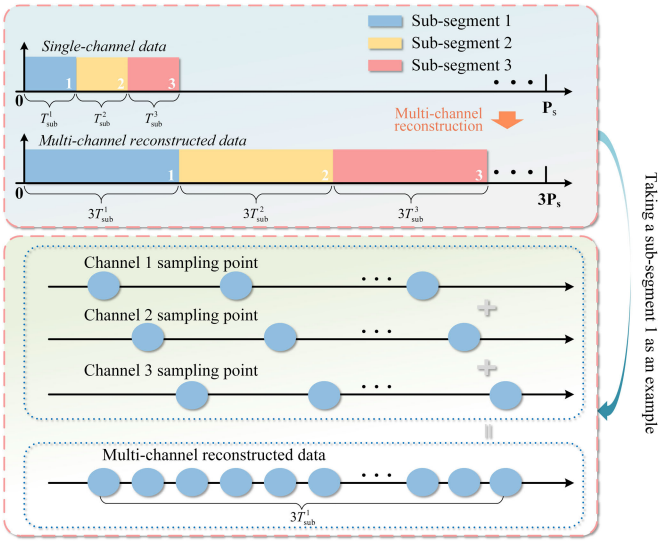


Fig. 3. Schematic of the proposed presegmented phase modulation idea before and after multichannel reconstruction (take 3 channels as an example, i.e., $M = 3$).

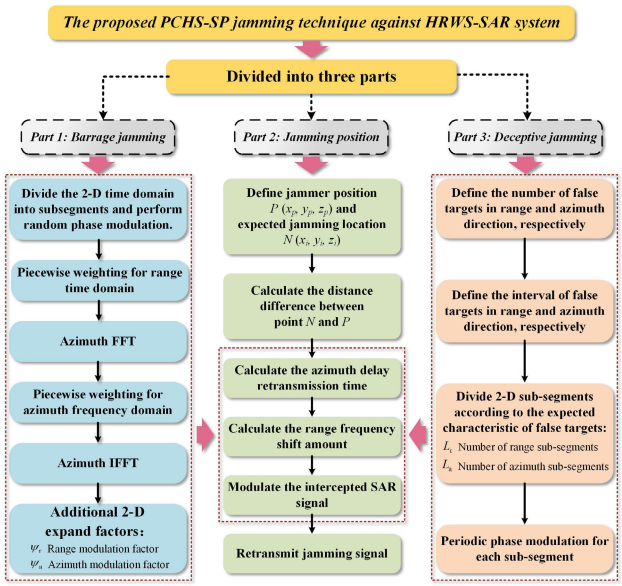


Fig. 4. Flowchart of the proposed PCHS-SP jamming technique.

The corresponding spectrum can be expressed as

$$\tilde{\mathbf{N}}(f_a) = \sum_{f_a^i=-B_n^a}^{B_n^a} \delta\left(f_a - \ddot{f}_a^i\right). \tag{12}$$

Define $\delta(f_a - \ddot{f}_a^{i_m})$ as the spectrum after folding $\delta(f_a - \ddot{f}_a^i - m*\mathbf{F}_p)$. Then, the spectrum after single-channel zero padding can be expressed as

$$\begin{aligned}\mathbf{S_N}(f_a) = &\sum_{\ddot{f}_a^i=-B_n^a}^{B_n^a} \delta\left(f_a - \ddot{f}_a^{i_0}\right) + \sum_{\ddot{f}_a^i=-B_n^a}^{B_n^a} \delta\left(f_a - \ddot{f}_a^{i_1}\right) \\ &+ \cdots + \sum_{\ddot{f}_a^i=-B_n^a}^{B_n^a} \delta\left(f_a - \delta\left(f_a - \ddot{f}_a^{i_{(M-1)}}\right)\right)\end{aligned} \tag{13}$$

Finally, the imaging results of NMMJ can be expressed as

$$\begin{aligned}&\mathbf{S_P}(\eta) \\ &= \sum_{i=1}^{M} \sum_{\ddot{f}_a^i=-B_n^a}^{B_n^a} \mathrm{g}\left(\ddot{f}_a^i\right)\left(\frac{B_a - \ddot{f}_a^{i_m}}{B_a}\right)\mathrm{sinc}\left[B_a\left(\eta - \frac{\ddot{f}_a^{i_m}}{K_a}\right)\right]\end{aligned} \tag{14}$$

where K_a represents the chirp rate in azimuth. From the above (14), it can be seen that for multifrequency jamming, that is, NMMJ, its imaging result is manifested as processing $M-1$ noise blocks in azimuth. In other words, it has a periodic repetitive jamming effect in the azimuth direction. In addition, it is not difficult to find that the reason why jamming recurs periodically in azimuth is that the spectrum of the Doppler-modulation jamming term will produce periodic repetition after multichannel reconstruction. This periodic jamming effect increases the likelihood that it can be detected and eliminated.

III. Proposed PCHS-SP Jamming Technique

According to Section II-B, the essence of the above problems is that the value of the jamming modulation term is the same in each channel, which resulted in the value of each sampling point being repeated $M-1$ times after the multichannel reconstruction. To avoid the above problem, this article introduces the idea of presegmented modulation. Fig. 3 shows the schematic before and after the multichannel reconstruction of the proposed presegmentation modulation idea. T_{sub}^1, T_{sub}^2, and T_{sub}^3 represent the data length (the number of sampling points) of subsegments 1, 2, and 3, respectively. $\mathbf{P_s}$ represents the total data length owned by a single channel. The lower half of Fig. 3 specifically shows the schematic before and after the multichannel reconstruction of subsegment 1. Clearly, after the multichannel processing, the data length of each subsegment in the proposed jamming modulation term is increased by a factor of M. Meanwhile, the differences between subsegments remain unchanged. Ultimately, the above problems are avoided. In addition, Appendix A presents the specific formula derivation to demonstrate how the proposed jamming method overcomes the limitations of multichannel reconstruction.

Specifically, this section proposes a 2-D precise controllable HRWS-SAR smart jamming method, that is, PCHS-SP, which can produce two jamming efforts: barrage and deceptive jamming. For barrage, 2-D expand factors are defined to achieve precise control of the jamming range. For deception, characteristics of false, including the number and amplitude, and so on, can be precisely controlled. Meanwhile, the 2-D precise controllability of the jamming position can also be achieved through the cooperation of the spatial and the signal domains. The detailed flowchart of the PCHS-SP method is shown in Fig. 4.

A. Barrage Jamming

The proposed barrage jamming implementation steps can be divided into three parts: random phase modulation, piecewise weighting, and additional expansion factors.

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1) Random Phase Modulation and Piecewise Weighting: This section introduces a segmented random phase modulation jamming method, which can produce a noise-like jamming effect in the 2-D mainlobe range of the point spread function and has low sidelobe characteristics. Without loss of generality, taking a point target as an example, the echo received by the jammer at the mth receiving channel can be expressed as

$$S_m^{J_1}(t_r,\eta) = p_r(t_r)p_a^m(\eta)\exp\left(-j\frac{2\pi f_0 R_m^J(\eta)}{c}\right) \times \exp\left[jK_r\left(t_r-\frac{R_m^J(\eta)}{c}\right)^2\right] \tag{15}$$

where K_r is the chirp rate of the LFM signal. p_r and p_a^m can be expressed as

$$p_r(t_r) = \omega_r\left(\frac{t_r - R_m^J(\eta)/c}{T_r}\right) \tag{16}$$

$$p_a^m(\eta) = \omega_a\left(\frac{\eta - x_j/v_a - D_m/v_a}{T_a}\right). \tag{17}$$

$\omega_r(\cdot)$ and $\omega_a(\cdot)$ represent the range and azimuth envelope curves, respectively. T_r is the pulse duration time. T_a represents the synthetic aperture time. $D_m = m\cdot d$ is the direct distance from the mth receiving channel to the transmitting channel. Here, $m = [-1, 0, 1]$ and d is the baseline difference between channels. $R_m^J(\eta)$ represents the two-way instantaneous slant range of the mth channel, which can be expressed as

$$R_m^J(\eta) = \sqrt{{R_0}^2 + \left(v_a\eta - x_j\right)^2} + \sqrt{{R_0}^2 + \left(v_a\eta - x_j - D_m\right)^2} \tag{18}$$

where R_0 represents the minimum slant range between the transmitting channel and the scattering point.

First, the first step is to divide the intercepted SAR signal into a series of subsegments, which can be achieved by using a rectangular window function of equal or unequal length. The next step is to perform random phase modulation for each segment. Phase modulation serves as the cornerstone of the proposed barrage jamming method, subject to the following constraints.

1) The time width of each subsegment can be any duration in the intercepted signal.
2) The phase value of each subsegment modulation can take any value on $[0, 2\pi]$, and the number of phase values should be greater than 3.

Ultimately, the random phase modulation jamming term can be expressed as

$$Q'(t_r,\eta) = \sum_{i_r=1}^{N_r}\left[\epsilon\left(t_r-\alpha_{i_r}^R\right) - \epsilon\left(t_r-\alpha_{i_r+1}^R\right)\right]\exp\left(j\phi_{i_r}\right) \cdot \sum_{i_a=1}^{N_a}\left[\epsilon\left(\eta-\alpha_{i_a}^A\right) - \epsilon\left(\eta-\alpha_{i_a+1}^A\right)\right]\exp\left(j\phi_{i_a}\right) \tag{19}$$

where $\epsilon(\cdot)$ represents the unit step function. $\alpha_{i_r}^R$ and $\alpha_{i_r+1}^R$ denote the starting and ending points of the i_rth subsegment in the range direction, respectively. $\alpha_{i_a}^A$ and $\alpha_{i_a+1}^A$ denote the starting and ending points of the i_ath subsegment in the azimuth direction, respectively. $\exp(j\phi_{i_r})$ and $\exp(j\phi_{i_a})$ represent the modulated random phases of the i_rth segment in the range direction and the i_ath segment in the azimuth direction, respectively. N_r and N_a represent the number of subsegments in the range and azimuth directions, respectively. In addition, it is worth noting that the subsegment division in the azimuth direction is performed against different pulses.

If the intercepted SAR signal is modulated according to (19), it will produce jamming effects resembling noise within the 2-D mainlobe range of the point spread function, but with relatively strong sidelobes. To minimize the effects of high sidelobes, this section utilizes a piecewise weighting technique, effectively focusing jamming energy within the mainlobe range. The weighting function can be implemented through window functions such as the cosine window, Chebyshev window, Gaussian window, and so on. In this section, we opt for the generalized cosine window function due to its simpler expression, facilitating easier control over the jamming effect.

For range direction, piecewise weighting can be directly performed in the time domain as it is independent of channels, which can be expressed as

$$Q_r'(t_r) = \sum_{i_r=1}^{N_r}\left[\beta + (1-\beta)\cos\left(2\pi\frac{t_r - \left(\alpha_{i_r+1}^R + \alpha_{i_r}^R\right)/2}{\alpha_{i_r+1}^R - \alpha_{i_r}^R}\right)\right] \tag{20}$$

where $\beta \in [0.5,\ 1]$ is the weighting coefficient and is used to control the degree of weighting in the range direction. At this point, the jamming signal generated by the mth channel can be expressed as

$$S_m^{J_2}(t_r,\eta) = S_m^{J_1}(t_r,\eta)\cdot Q'(t_r,\eta)\cdot Q_r'(t_r). \tag{21}$$

For the azimuth direction, to overcome the azimuth periodic repetition caused by multichannel reconstruction processing, piecewise weighting is performed in the RD domain. The piecewise weighting jamming term in the azimuth direction can be expressed as

$$Q_a'(f_a) = \sum_{i_a=1}^{N_a}\left[\gamma + (1-\gamma)\cos\left(2\pi\frac{f_a - \left(\alpha_{i_a+1}^A + \alpha_{i_a}^A\right)/2}{\alpha_{i_a+1}^A - \alpha_{i_a}^A}\right)\right] \tag{22}$$

where $\gamma \in [0.5,\ 1]$ is the weighting coefficient and is used to control the degree of weighting in the azimuth direction. Finally, the generated 2-D jamming signal can be expressed as

$$S_m^{J_3}(t_r,\eta) = \mathrm{IFFT}\left\{\mathrm{FFT}\left[S_m^{J_2}(t_r,\eta)\right]\cdot Q_a'(f_a)\right\}. \tag{23}$$

After multichannel imaging processing, the above jamming modulation items can form a jamming effect similar to noise within the 2-D mainlobe range of the point spread function. Simultaneously, the sidelobes can be controlled by adjusting

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the number of subsegments and weighting coefficients to meet complex battlefield requirements.

2) 2-D Expansion Factors: Although $S_m^{J_3}$ can produce a certain degree of barrage jamming effect, it is confined to the local range only. To achieve precise 2-D control over the jamming range, this section introduces range and azimuth expansion factors and establishes a parametric relationship between 2-D expansion factors and the jamming range. Noteworthy, the expansion factors mentioned in this section are implemented based on the chirp rate error.

Assuming a chirp rate error K_r^e exists in the jamming signal after modulation in the range direction, the generated jamming signal in the range direction can be represented as

$$S_m^{J_4}(t_r) = p_r(t_r) exp\left[j\left(K_r + K_r^e\right)\left(t_r - \frac{R_m^J(\eta)}{c}\right)^2\right] Q_r''(t_r) \tag{24}$$

where $Q_r''(t_r)$ can be expressed as

$$Q_r''(t_r) = \sum_{i_r=1}^{N_r}\left[\epsilon\left(t_r - \alpha_{i_r}^R\right) - \epsilon\left(t_r - \alpha_{i_r+1}^R\right)\right]\exp\left(j\phi_{i_r}\right) \cdot \left[\beta + (1-\beta)\cos\left(2\pi\frac{t_r - \left(\alpha_{i_r+1}^R - \alpha_{i_r}^R\right)/2}{\alpha_{i_r+1}^R - \alpha_{i_r}^R}\right)\right]. \tag{25}$$

For SAR imaging technology using LFM signals, the chirp rate error will inevitably lead to the mainlobe broadening of the point spread function. Therefore, after PC, $S_m^{J_4}(t_r)$ no longer exhibits the shape of a sinc function, but has a flat peak (FP) shape. In addition, the existence of $Q_r''(t_r)$ will further destroy the correlation between the jamming signal and the matched filter, which causes the mainlobe to expand. However, compared with the degree of broadening caused by the chirp rate error, the mainlobe broadening caused by $Q_r''(t_r)$ can usually be ignored. Against this part, we have discussed it in detail in previous work [32].

To facilitate the discussion of the jamming range, this section describes FP caused by the chirp rate error as a rectangular window function. Meanwhile, the parameter relationship between the chirp rate error and the FP width is established. First, for the range direction, this relationship can be described as

$$\mathbf{I}_R^+\left(t_r, \eta, K_r^e\right) = \varpi\left(\frac{t_r - R_m^J(\eta)/c}{K_r^e T_r/\left(K_r + K_r^e\right)}\right) \cdot \exp\left[-j\pi\frac{\left(K_r + K_r^e\right)K_r}{K_r^e}\left(t_r - \frac{R_m^J(\eta)}{c}\right)^2\right] \tag{26}$$

where $\varpi(\cdot)$ indicates a rectangular window function centered on 0. Noteworthy, $\mathbf{I}_R^+(t_r, \eta, K_r^e)$ is an expression established based on $K_r^e > 0$, $K_r^e \in \mathbf{R}^+$. If $K_r^e < 0$, the range direction FP width of $S_m^{J_4}(t_r)$ after PC can be described as

$$\mathbf{I}_R^-\left(t_r, \eta, K_r^e\right) = \varpi\left(\frac{t_r - R_m^J(\eta)/c}{K_r^e T_r/K_r}\right) \cdot \exp\left[-j\pi\frac{\left(K_r + K_r^e\right)K_r}{K_r^e}\left(t_r - \frac{R_m^J(\eta)}{c}\right)^2\right]. \tag{27}$$

To achieve precise control over the range direction to the jamming range, the range expansion factor can be defined as follows:

$$\psi_r = \frac{K_r + K_r^e}{K_r^e}K_r. \tag{28}$$

Hence, the parameterized relationship between the jamming range of the range direction and the expansion factor can be derived as

$$\Delta U_r = \begin{cases} \left|\frac{cB_r}{2\psi_r}\right| & K_r^e \in \mathbf{R}^+ \\ \left|\frac{cB_r + 2\psi_r K_r}{2\psi_r}\right| & K_r^e \in \mathbf{R}^-. \end{cases} \tag{29}$$

For the azimuth direction, R_m^J can be approximated as

$$R_m^J(\eta) = 2R_0 - \frac{v_a x_j + D_m}{R_0}\eta + \frac{v_a^2}{R_0}\eta^2 + \frac{2x_j D_m + D_m^2 + x_j^2}{2R_0}. \tag{30}$$

According to (30), the jamming signal of the mth channel in the azimuth direction can be re-expressed as

$$S_m^{J_5}(t_r) = p_a^m(\eta)\exp\left[j\frac{2\pi\left(v_a x_j + D_m\right)f_0\eta}{cR_0}\right]\exp\left(-j\frac{2\pi v_a^2 f_0\eta^2}{cR_0}\right) \cdot \exp\left[-j\frac{\pi v\left(2x_j D_m + D_m^2 + x_j^2\right)f_0}{cR_0}\right]\exp\left[-j\frac{4\pi R_0 f_0}{c}\right]. \tag{31}$$

The linear phase term determines the peak position of the signal pulse after PC. The constant phase term is typically negligible and can be ignored. Therefore, the HRWS-SAR azimuth mainlobe width is determined by the quadratic phase term $\exp(-j(2\pi v_a^2 f_0\eta^2/cR_0))$. Fortunately, it can be regarded as an LFM signal with a chirp rate of $K_a = (2v_a^2/\lambda R_0)$. By analogy to the range direction, azimuth chirp rate error is further employed to achieve control over the jamming range in the azimuth direction. Correspondingly, the rectangular window function is used to characterize the PC output with azimuth chirp rate error K_a^e jamming signal, its FP width can be expressed as

$$\widehat{\mathbf{I}}\left(t_r, \eta, K_a^e\right) = \begin{cases} \varpi\left(\frac{\eta - x_j/v_a - D_m/v_a}{K_a^e T_a/\left(K_a + K_a^e\right)}\right)\Phi(\eta) & K_a^e \in \mathbf{R}^+ \\ \varpi\left(\frac{\eta - x_j/v_a - D_m/v_a}{K_a^e T_a/K_a}\right)\Phi(\eta) & K_a^e \in \mathbf{R}^- \end{cases} \tag{32}$$

where $\Phi(\eta)$ can be expressed as

$$\Phi(\eta) = \exp\left[j\pi\frac{\left(K_a + K_a^e\right)K_a\left(\eta - x_j/v_a - D_m/v_a\right)^2}{K_a^e}\right]. \tag{33}$$

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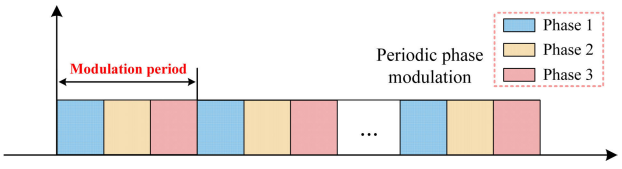


Fig. 5. Schematic of periodic phase modulation.

Define the expansion factor in the azimuth direction as ψ_a, which can be expressed as

$$\psi_a = \frac{K_a + K_a^e}{K_a^e} K_a. \tag{34}$$

Finally, the parametric relationship between the azimuth spreading factor and the jamming range can be expressed as

$$\Delta U_a = \begin{cases} \left|\dfrac{c v_a^2 T_a}{\lambda R_0 \psi_a}\right| & K_a^e \in \mathbf{R}^+ \\ \left|\dfrac{c v_a^2 T_a + \lambda R_0 \psi_a K_a}{\lambda R_0 \psi_a}\right| & K_a^e \in \mathbf{R}^-. \end{cases} \tag{35}$$

By utilizing (29) and (35), it becomes straightforward to calculate the 2-D expansion factor corresponding to the predefined jamming range. Furthermore, we can proceed to determine the necessary 2-D chirp rate. This lays the foundation for achieving precise 2-D control over the jamming range.

B. Deceptive Jamming

Deceptive jamming is achieved by imposing specific restrictions against the phase values of each subsegmented modulation, that is, periodic phase modulation, building upon the foundation of barrage jamming. Fig. 5 shows the schematic of periodic phase modulation.

Phase modulation is also central to the implementation of deceptive jamming. However, unlike barrage jamming, it introduces some new unique limitations.

1) The intercepted SAR signal is partitioned into multiple subsegments in the 2-D time domain, with each subsegment having an equal time duration.
2) The modulation phase values have specific regulations, that is, appear periodically in repetition.

Compared to barrage jamming, deceptive jamming allows for simplified representation using impulse functions $\delta(\cdot)$ due to the equal duration of each subsegment in the deceptive jamming scheme. Assuming the range direction is divided into L_r subsegments, the jamming modulation term for deceptive jamming can be represented as

$$\mathbf{J}_r(t_r) = \sum_{l_r}^{L_r} \mathrm{rect}\left(\frac{t_r - 2R_m^J(\eta)/c}{\tau_r}\right) \cdot \delta\left(t_r - l_r\tau_r - \frac{2R_m^J(\eta)}{c}\right)\exp\left(j\theta_{l_r}^R\right) \tag{36}$$

where $\tau_r = T_r/L_r$ represents the time duration of each subsegment. $\exp(j\theta_{l_r}^R)$ represents the modulation phase value in the range direction, which repeats periodically. Assuming the azimuth direction is divided into L_a subsegments of equal duration, each with a time length of τ_a, the modulation deceptive jamming term in the azimuth direction can be represented as

$$\mathbf{J}_a(\eta) = \sum_{l_a}^{L_a} \mathrm{rect}\left(\frac{\eta - x_j/v_a - D_m/v_a}{\tau_a}\right) \cdot \delta\left(\eta - l_a\tau_a - v_a x_j - D_m\right)\exp\left(j\theta_m^A\right). \tag{37}$$

$\exp(j\theta_m^A)$ represents the modulation phase value in the azimuth direction, which also repeats periodically. Ultimately, the deceptive jamming signal received by the mth channel can be represented as

$$S_m^{J_6}(t_r, \eta) = S_m^{J_1} \cdot \mathbf{J}_r(t_r) \cdot \mathbf{J}_a(\eta). \tag{38}$$

Based on (38), it can be deduced that the primary parameters influencing the PC result in $S_m^{J_6}(t_r, \eta)$ include subsegment length, phase values, and so on. This offers greater flexibility and controllability for the implementation of deceptive jamming.

C. 2-D Precise Control of Jamming Position

To avoid periodic repetition of the orientation term, this section introduces a method of jamming modulation that combines spatial and signal domain modulation, which can enable precise and controllable 2-D jamming positions. Specifically, this article specifies the deployment position of the jamming device and the transmission time of the jamming signal from a spatial perspective. By utilizing delayed retransmission of the jamming signal, control over the azimuth jamming location is achieved. Furthermore, the jamming signal in the range direction, since it does not participate in the multichannel reconstruction processing of HRWS-SAR, can be easily achieved by simply adding a linear phase.

Assuming the predetermined jamming position is $N(x_i, y_i, z_i)$, the 2-D time difference between it and the position of the jammer can be represented as

$$\begin{cases} \Delta_{\mathrm{Azi}} = \left|\dfrac{x_p - x_i}{v_a}\right| & \text{Azimuth} \\ \Delta_{\mathrm{Ran}} = \dfrac{y_p - y_i}{c} & \text{Range.} \end{cases} \tag{39}$$

For an HRWS-SAR working in a "stop-and-go" mode, the azimuthal delayed retransmission needs to be based on the PRF, which can be represented as

$$\widetilde{\Delta}_{\mathrm{Azi}} = \lceil(\Delta_{\mathrm{Azi}} - \mu) \cdot \mathbf{F}_p\rceil \tag{40}$$

where μ represents the working delay of the jammer. After undergoing delayed retransmission for $\widetilde{\Delta}_{\mathrm{Azi}}$ pulse repetition times, the generated jamming signal can achieve precise control over the azimuthal jamming location.

Based on the time–frequency coupling characteristics of the LFM signal, the linear phase modulation, that is, the frequency shift item, can be represented as

$$\widetilde{\Delta}_{\mathrm{Ran}} = \exp(j2\pi K_r t_r \Delta_{\mathrm{Ran}}). \tag{41}$$

By manipulating the jamming term as indicated in modulation (41), precise control over the direction of the range jamming location can also be achieved.

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IV. Jamming Parameter and System Error Analysis

Unlike the conventional single-channel SAR imaging mechanism, HRWS-SAR requires multichannel reconstruction before traditional imaging algorithms can be used to obtain high-resolution images. First, the HRWS-SAR echo without jamming after multichannel reconstruction can be described as

$$\widehat{S}(t_r,\widehat{\eta}) = p_r(t_r)p_a(\widehat{\eta})\exp\left(-j\frac{2\pi f_0 R_J(\widehat{\eta})}{c}\right) \cdot \exp\left[jK_r\left(t_r-\frac{R_J(\widehat{\eta})}{c}\right)^2\right] \tag{42}$$

where $\widehat{\eta}$ represents the azimuth sampling time after multichannel reconstruction sampling. $R_J(\widehat{\eta})$ is the instantaneous slant range from the jammer to the phase center of the HRWS-SAR receiving antenna after multichannel reconstruction. Subsequently, it is only necessary to consider the effect of the jamming modulation term after multichannel reconstruction on HRWS-SAR imaging.

A. Barrage Jamming Parameter Analysis

From Section III-A, it can be observed that the parameters affecting barrage jamming mainly include the number of subsegments, weighting coefficient, and expansion factor.

1) Number of Subsegments: From Fig. 3, it can be observed that the number of subsegments and the differences between subsegments remain unchanged after multichannel reconstruction. Therefore, after multichannel reconstruction processing, the jamming signal generated based on piecewise random phase modulation can be represented as

$$\begin{aligned}\widehat{S}_{\mathbf{J}}^{1}(t_r,\widehat{\eta}) = \widehat{S}(t_r,\widehat{\eta}) &\cdot\sum_{i_r=1}^{N_r}\left[\epsilon\left(t_r-\alpha_{i_r}^{R}\right)-\epsilon\left(t_r-\alpha_{i_r+1}^{R}\right)\right]\exp\left(j\phi_{i_r}\right)\\ &\cdot\sum_{i_a=1}^{N_a}\left[\epsilon\left(\widehat{\eta}-\widehat{\alpha}_{i_a}^{A}\right)-\epsilon\left(\widehat{\eta}-\widehat{\alpha}_{i_a+1}^{A}\right)\right]\exp\left(j\phi_{i_a}\right)\end{aligned} \tag{43}$$

where $\widehat{\alpha}_{i_a}^{A}$ and $\widehat{\alpha}_{i_a+1}^{A}$ denote the starting and ending points of the i_ath subsegment after multichannel reconstruction, respectively. Using the RD algorithm as the imaging algorithm, the 2-D PC result after range cell migration correction (RCMC) can be expressed as

$$\begin{aligned}&\widetilde{S}_1(t_r,\widehat{\eta})\\ &=\sum_{i_r=1}^{N_r}\Delta\alpha_{i_r}^{R}\mathrm{sinc}\left(K_r\widetilde{t}_r\Delta\alpha_{i_r}^{R}\right)\exp\left(j\phi_{i_r}\right)\\ &\quad\cdot\exp\left(-j\pi K_r\widetilde{t}_r{}^2\right)\exp\left[j\pi K_r\widetilde{t}_r\left(\alpha_{i_r+1}^{R}+\alpha_{i_r}^{R}\right)\right]\\ &\quad\cdot\sum_{i_a=1}^{N_a}\Delta\widehat{\alpha}_{i_a}^{A}\mathrm{sinc}\left[K_a\widetilde{\eta}\Delta\widehat{\alpha}_{i_a}^{A}\right]\exp\left(j\phi_{i_a}\right)\\ &\quad\cdot\exp\left(-j\pi K_a\widetilde{\eta}\ {}^2\right)\exp\left[j\pi K_a\widetilde{\eta}\left(\widehat{\alpha}_{i_a+1}^{A}+\widehat{\alpha}_{i_a}^{A}\right)\right]\end{aligned} \tag{44}$$

where $\widetilde{t}_r = t_r - 2R_J(\widehat{\eta})/c$ and $\widetilde{\eta} = \widehat{\eta} - x_j/v_a$. $\Delta\alpha_{i_r}^{R} = \alpha_{i_r+1}^{R} - \alpha_{i_r}^{R}$ and $\Delta\widehat{\alpha}_{i_a}^{A} = \widehat{\alpha}_{i_a+1}^{A} - \widehat{\alpha}_{i_a}^{A}$ are the segment lengths in the range and azimuth directions after multichannel reconstruction processing, respectively. Particularly, if all subsegments have the same length, that is, they are evenly divided, then (44) can be further simplified to

$$\begin{aligned}&\widetilde{S}_2(t_r,\widehat{\eta})\\ &=\frac{T_r}{N_r}\mathrm{sinc}\left(K_r\widetilde{t}_r\frac{T_r}{N_r}\right)\exp\left[-j\pi K_r\widetilde{t}_r\left(\frac{N_r+1}{2N_r}\right)\right]\\ &\quad\cdot\exp\left(-j\pi K_r\widetilde{t}_r{}^2\right)\sum_{i_r=1}^{N_r}\exp\left(j\phi_{i_r}\right)\exp\left[j\pi K_rT_r\widetilde{t}_r\left(\frac{i_r}{N_r}\right)\right]\\ &\quad\cdot\frac{T_a}{N_a}\mathrm{sinc}\left(K_a\widetilde{\eta}\frac{T_a}{N_a}\right)\exp\left[-j\pi K_a\widetilde{\eta}\left(\frac{N_a+1}{2N_a}\right)\right]\\ &\quad\cdot\exp\left(-j\pi K_a\widetilde{\eta}_a{}^2\right)\sum_{i_a=1}^{N_a}\exp\left(j\phi_{i_a}\right)\exp\left[j\pi K_aT_a\widetilde{\eta}\left(\frac{i_a}{N_a}\right)\right].\end{aligned} \tag{45}$$

From (44) and (45), it can be inferred that if the subsegment lengths remain equal, $\widetilde{S}_2(t_r,\widehat{\eta})$ exhibits an envelope form resembling a 2-D sinc function. This concentrates the energy mainly in the mainlobe. However, due to the influence of the summation term against the random phase modulation, the envelope curve of the 2-D sinc function is no longer smooth in this case. Instead, it exhibits a certain degree of randomness, that is, noise-like jamming effects will occur within the mainlobe of the 2-D sinc function. This lays the foundation for barrage jamming.

Furthermore, the amplitude coefficients of $\widetilde{S}_2(t_r,\widehat{\eta})$ 2-D sinc function in the range and azimuth directions are T_r/N_r and T_a/N_a, respectively. For echo without jamming, the amplitude coefficients of the real target are T_r and T_a, respectively. Thus, it can be concluded that the amplitude coefficients of the jamming signal after subsegment random phase modulation are $1/N_r$ and $1/N_a$ times that of the true target, respectively. In other words, the larger the number of segments, the smaller the amplitude of $\widetilde{S}_2(t_r,\widehat{\eta})$. Furthermore, the mainlobe widths of $\widetilde{S}_2(t_r,\widehat{\eta})$ are N_r/B_r and N_a/B_r, respectively. B_a and B_r represent the range bandwidth and Doppler bandwidth, respectively. Therefore, the mainlobe width of $\widetilde{S}_2(t_r,\widehat{\eta})$ is directly proportional to the number of subsegments, that is, the larger the number of subsegments, the wider the mainlobe width of $\widetilde{S}_2(t_r,\widehat{\eta})$. In summary, the number of subsegments provides more controllability for barrage jamming.

2) Weighting Coefficient: Since the azimuthal piecewise weighting is performed in the RD domain, the range and azimuth directions will be discussed separately for ease of analysis. First, after segmented weighting, the jamming signal in the range direction can be represented as

$$\begin{aligned}&\widetilde{S}_J^2(t_r,\widehat{\eta})\\ &=p_r(t_r)\exp\left[jK_r\left(t_r-\frac{R_J(\widehat{\eta})}{c}\right)^2\right]\end{aligned}$$

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$$\cdot \sum_{i_r=1}^{N_r} \left[\epsilon\left(t_r - \alpha_{i_r}^R\right) - \epsilon\left(t_r - \alpha_{i_r+1}^R\right)\right] \exp\left(j\phi_{i_r}\right) \cdot \left[\beta + (1-\beta)\cos\left(2\pi \frac{t_r - \left(\alpha_{i_r+1}^R + \alpha_{i_r}^R\right)/2}{\alpha_{i_r+1}^R - \alpha_{i_r}^R}\right)\right]. \quad (46)$$

The PC result of cosine amplitude weighting comprises three sinc functions with different amplitudes and offsets. The sinc function with offset is located between the mainlobe and the first sidelobe of the function without offset, thereby resulting in a new sinc function with low sidelobes [30]. Equation (48), as shown at the bottom of the next page, is the PC output result of $\widetilde{S}_3(\widehat{\eta})$. From (48), it is evident that the result of range PC after piecewise weighting can be viewed as a weighted sum of sinc functions with different weighting coefficients and offsets in the range direction for each subsegment. The sidelobes of the without offset sinc function are absorbed to varying degrees into the mainlobe of the offset sinc function. Ultimately, this will form a new sinc function with low sidelobes. Therefore, the sidelobes of barrage jamming signals in the range direction can be controlled by β.

For the azimuth direction, (45) has already provided the result of PC after segmented random phase modulation. Therefore, we only need to consider the impact of the piecewise weighting term on the PC results after multichannel reconstruction.

To simplify the analysis, let us first consider the piecewise modulation jamming term in the azimuth direction for the i_a segment before multichannel reconstruction, which can be represented as

$$\mathbf{J}_{i_a}^b(f_a) = \gamma + (1-\gamma)\cos\left(2\pi \frac{f_a - \left(\alpha_{i_a+1}^A + \alpha_{i_a}^A\right)/2}{\alpha_{i_a+1}^A - \alpha_{i_a}^A}\right). \quad (47)$$

If $\mathbf{J}_{i_a}^b(f_a)$ is inverse Fourier transformed, it can be expressed as

$$\begin{aligned}\mathbf{J}_{i_a}^b(\eta) &= \gamma\delta(\eta) \\ &+ \frac{1-\gamma}{2}\left[\exp\left(-j\pi \frac{\alpha_{i_a+1}^A + \alpha_{i_a}^A}{\alpha_{i_a+1}^A - \alpha_{i_a}^A}\right)\delta\left(\eta + \frac{2\pi}{\alpha_{i_a+1}^A - \alpha_{i_a}^A}\right)\right] \\ &+ \frac{1-\gamma}{2}\left[\exp\left(j\pi \frac{\alpha_{i_a+1}^A + \alpha_{i_a}^A}{\alpha_{i_a+1}^A - \alpha_{i_a}^A}\right)\delta\left(\eta - \frac{2\pi}{\alpha_{i_a+1}^A - \alpha_{i_a}^A}\right)\right].\end{aligned} \quad (49)$$

From (49), it can be inferred that before multichannel reconstruction, the time-domain expression of the piecewise weighted term for segment i_a is formed by $\delta(\cdot)$ function with different weighting coefficients γ and offsets. By the convolution property of the impulse function, we can conclude that frequency-domain piecewise weighting has the same effect as time-domain piecewise weighting. The next step is to consider the influence of multichannel reconstruction for the piecewise weighting jamming term.

According to Section II-B, it can be seen that the discretization of $\mathbf{J}_{i_a}^b(\eta)$ after multichannel reconstruction can be expressed as

$$\widehat{\mathbf{J}}_{i_a}^b(\widehat{\eta}) = \mathbb{J}_{i_a}^b(\widehat{\eta}) + \mathbb{J}_{i_a}^b(\widehat{\eta}-1) + , \ldots, + \mathbb{J}_{i_a}^b(\widehat{\eta}-M+1) \quad (50)$$

where $\mathbb{J}_{i_a}^b(l)$ can be expressed as

$$\mathbb{J}_{i_a}^b(\widehat{\eta}) = 0, 0, \ldots, \underbrace{\frac{1-\gamma}{2}, \ldots, \gamma, \ldots, \frac{1-\gamma}{2}}_{\lceil \frac{2\pi \cdot M}{\widehat{\alpha}_{i_a+1}^A - \widehat{\alpha}_{i_a}^A} \rceil}, \ldots, 0 \quad (51)$$

where $\lceil\cdot\rceil$ represents rounding up. Therefore, the i_ath jamming signal after piecewise weighting can be further expressed as

$$\widetilde{S}_{i_a}^{\mathbf{J}}(t_r, \widetilde{\eta}) = \left\{ p_a(\widehat{\eta}) \exp\left(-j\frac{2\pi f_0 R_J(\widehat{\eta})}{\lambda}\right) \exp\left(j\phi_{i_a}\right) \cdot \left[\epsilon\left(\widehat{\eta} - \widehat{\alpha}_{i_a}^A\right) - \epsilon\left(\widehat{\eta} - \widehat{\alpha}_{i_a+1}^A\right)\right]\right\} \otimes \widehat{\mathbf{J}}_{i_a}^b(\widehat{\eta}). \quad (52)$$

Subsequently, the PC of $\widetilde{S}_{i_a}^{\mathbf{J}}(t_r, \widetilde{\eta})$ can be expressed as

$$\widetilde{S}_{i_a}^{\mathbf{c}}(\widehat{\eta}) = \widetilde{S}_{i_a}^{o}(\widehat{\eta}) + \widetilde{S}_{i_a}^{o}(\widehat{\eta}-1) + \cdots \widetilde{S}_{i_a}^{o}(\widehat{\eta}-M+1) \quad (53)$$

where $S_{i_a}^{o}(\widehat{\eta})$ can be expressed as (54), as shown at the bottom of page 12.

By comparing (48) and (54), it is easy to see that they have similar function forms, that is, composed of sinc functions with different weighting coefficients and offsets. This results in $\widetilde{S}_{i_a}^{o}(\widehat{\eta})$ having low sidelobe characteristics. Furthermore, $\widetilde{S}_{i_a}^{\mathbf{c}}(\widehat{\eta})$ can be regarded as a new sinc function composed of M low sidelobe sinc functions with different offsets. The M sinc functions composing $\widetilde{S}_{i_a}^{o}(\widehat{\eta})$ are closely spaced, differing only by one sampling point. The closely spaced and same waveform sinc functions in the distance will make them difficult to distinguish in the time domain, appearing as a single sinc function, but the width of the mainlobe will increase significantly. Therefore, $\widetilde{S}_{i_a}^{\mathbf{c}}(\widehat{\eta})$ still has the characteristics of low sidelobe.

In conclusion, if piecewise weighting is performed in the azimuth frequency domain, a low sidelobe PC output result can similarly be obtained for each subsegmentation. Like the range direction, the sum of N_a low sidelobe sinc functions still exhibit low sidelobe characteristics. Therefore, the azimuth sidelobe can also be suppressed by performing piecewise weighting in the RD domain.

3) Expansion Factors: The modulation factor primarily involves modulating the intercepted SAR signal with chirp rate errors to expand the jamming range further. Equations (29) and (35) describe the parametric relationship between the 2-D expansion factor and the jamming range. Specifically, according to the preset jamming range, its 2-D corresponding modulation factor can be easily solved. Furthermore, the corresponding chirp rate error is also easily solved. Finally, Fig. 6 shows the relationship between the chirp rate error and barrage jamming range (it is important to note that the jamming range refers to the mainlobe width when it decreases by 7.5 dB) [34]. This parameterized relationship provides the foundation for achieving precise 2-D control over the jamming range.

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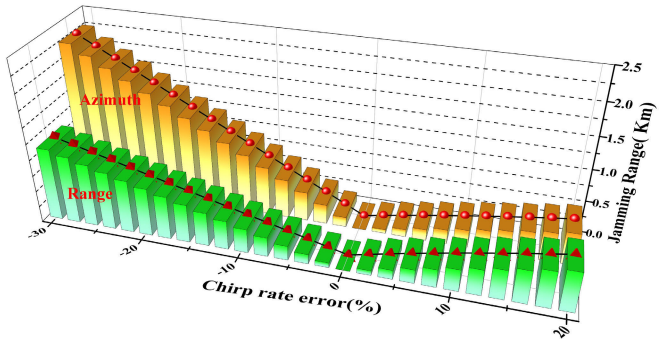


Fig. 6. Diagram of the relationship between the chirp rate error and the jamming range.

B. Deceptive Jamming Parameter Analysis

For deceptive jamming, the characteristics of false targets, including, quantity, amplitude, and so on, are often the focus of attention. This section will analyze each of them in turn.

Similar to barrage jamming, the number of subsegments and the differences between subsegments remain unchanged after the multichannel reconstruction of deceptive jamming. Therefore, the deceptive jamming signal after multichannel reconstruction can be expressed as

$$\widehat{S}_{\mathbf{J}}^{3}(t_r,\widehat{\eta}) = \widehat{S}(t_r,\widehat{\eta})\cdot \mathbf{J}_r(t_r)\cdot \widehat{\mathbf{J}}_a(\widehat{\eta}) \tag{55}$$

where $\widehat{\mathbf{J}}_a(\widehat{\eta})$ represents the equation of $\mathbf{J}_a(\eta)$ after multichannel reconstruction processing. Using the RD algorithm, the PC result of $\widehat{S}_{\mathbf{J}}^{3}(t_r,\widehat{\eta})$ can be expressed as

$$\begin{aligned}&\widetilde{S}_3(t_r,\widehat{\eta})\\ &=\sum_{-A}^{A} a_{\overline{\mathbf{n}}_r}\operatorname{sinc}\left[T_r\left(K_r\widetilde{t}_r+\frac{\overline{\mathbf{n}}_r}{\mathbf{b}_r\tau_r}\right)\left(1-\frac{|\widetilde{t}_r|}{T_r}\right)\right]\left(1-\frac{|\widetilde{t}_r|}{T_r}\right)\\ &\quad\cdot\sum_{-B}^{B} b_{\overline{\mathbf{n}}_a}\operatorname{sinc}\left[T_a\left(K_a\widetilde{\eta}+\frac{\overline{\mathbf{n}}_a}{\mathbf{b}_a\widehat{\tau}_a}\right)\left(1-\frac{|\widetilde{\eta}|}{T_a}\right)\right]\left(1-\frac{|\widetilde{\eta}|}{T_a}\right)\end{aligned} \tag{56}$$

where $A=\lfloor B_r\mathbf{b}_r\tau_r/2\rfloor$, $\overline{\mathbf{n}}_r=\pm1,\pm2,\ldots,\pm A$ represents the order of the false target in the range direction. $B=\lfloor B_a\mathbf{b}_a\widehat{\tau}_a/2\rfloor$, $\overline{\mathbf{n}}_a=\pm1,\pm2,\ldots,\pm B$ represents the order of the false target in the azimuth direction. B_r and B_a represent the range and azimuth bandwidth, respectively. $\widehat{\tau}_a$ represents the time duration of each subsegment in the azimuth direction after multichannel reconstruction. $\mathbf{b}_r$ and $\mathbf{b}_a$ represent the number of subsegments in a repeating period in the range and azimuth directions, respectively. $a_{\overline{\mathbf{n}}_r}$ and $b_{\overline{\mathbf{n}}_a}$ can be expressed as

$$\begin{cases} a_{\overline{\mathbf{n}}_r}=\dfrac{\operatorname{sinc}(\overline{\mathbf{n}}_r/\mathbf{b}_r)}{\mathbf{b}_r}\displaystyle\sum_{l_r=0}^{\mathbf{b}_r-1}\exp\left(j\theta_{l_r}^{R}\right)\exp\left(\frac{j2\pi\overline{\mathbf{n}}_r l_r}{\mathbf{b}_r}\right)\\ b_{\overline{\mathbf{n}}_a}=\dfrac{\operatorname{sinc}(\overline{\mathbf{n}}_a/\mathbf{b}_a)}{\mathbf{b}_a}\displaystyle\sum_{l_a=0}^{\mathbf{b}_a-1}\exp\left(j\theta_{l_a}^{A}\right)\exp\left(\frac{j2\pi\overline{\mathbf{n}}_a l_a}{\mathbf{b}_a}\right).\end{cases} \tag{57}$$

1) Amplitude of the False Target: From (56), the amplitude of the false target at $(\overline{\mathbf{n}}_r,\overline{\mathbf{n}}_a)$ can be expressed as

$$\Delta\mathbf{Q}(\overline{\mathbf{n}}_r,\overline{\mathbf{n}}_a)=\left|a_{\overline{\mathbf{n}}_r}b_{\overline{\mathbf{n}}_a}\left(\frac{j2\pi\overline{\mathbf{n}}_r l_r}{\mathbf{b}_r}\right)\left(\frac{j2\pi\overline{\mathbf{n}}_a l_a}{\mathbf{b}_a}\right)\right|. \tag{58}$$

If the transmission signal and jamming modulation signal remain unchanged, $((j2\pi\overline{\mathbf{n}}_r l_r/\mathbf{b}_r))((j2\pi\overline{\mathbf{n}}_a l_a/\mathbf{b}_a))$ is typically constant. Therefore, the amplitude of the false target at $(\overline{\mathbf{n}}_r,\overline{\mathbf{n}}_a)$ mainly depends on $a_{\overline{\mathbf{n}}_r}b_{\overline{\mathbf{n}}_a}$. This provides a basis for achieving controllable false target amplitude.

2) Position of the False Target: From (56), the position of the false target at $(\overline{\mathbf{n}}_r,\overline{\mathbf{n}}_a)$ can be expressed as

$$\Delta\mathbf{R}(\overline{\mathbf{n}}_r,\overline{\mathbf{n}}_a)=\begin{cases}(\overline{\mathbf{n}}_r c/2K_a\mathbf{b}_r\tau_r+R_0) & \text{Range}\\ \left(\overline{\mathbf{n}}_a v_a/2K_a\mathbf{b}_a\widehat{\tau}_a\right) & \text{Azimuth.}\end{cases} \tag{59}$$

To distinguish different false targets well on SAR images, the intervals between false targets should meet the following conditions:

$$\begin{cases} c/2K_a\mathbf{b}_r\tau_r>\rho_r & \text{Rnage}\\ v_a/2K_a\mathbf{b}_a\widehat{\tau}_a>\rho_a & \text{Azimuth}\end{cases} \tag{60}$$

where ρ_r and ρ_a represent range and azimuth resolution, respectively.

3) Number of False Targets: If $\overline{\mathbf{n}}_r/\mathbf{b}_r\tau_r>B_r$ and $\overline{\mathbf{n}}_a/\mathbf{b}_a\widetilde{\tau}_a>B_a$, since the corresponding spectrum exceeds its frequency range, the corresponding false targets do not appear in the imaging results. In addition, the symmetry of the periodic signal makes the frequency components also symmetrical. Therefore, the total number of false targets is $E=(2A+1)\cdot(2B+1)$.

Based on the above analysis, controlling factors such as the number of segments, modulation period, and so on, enable the manipulation of the characteristics of false targets, thereby providing greater controllability and flexibility for the implementation of deceptive jamming.

$$\begin{aligned}&\widetilde{S}_3(\widehat{t}_r)\\ &=\sum_{i_r=1}^{N_r}\Delta\alpha_{i_r}^{R}\operatorname{sinc}\left(K_r\widetilde{t}_r\Delta\alpha_{i_r}^{R}\right)\exp\left(j\phi_{i_r}\right)\exp\left(-j\pi K_r\widetilde{t}_r{}^2\right)\exp\left[j\pi K_r\widetilde{t}_r\left(\alpha_{i_r+1}^{R}+\alpha_{i_r}^{R}\right)\right]\\ &\quad+\frac{1-\beta}{2}\sum_{i_r=1}^{N_r}\Delta\alpha_{i_r}^{R}\operatorname{sinc}\left[\left(K_r\widetilde{t}_r+\frac{1}{\Delta\alpha_{i_r}^{R}}\right)\Delta\alpha_{i_r}^{R}\right]\exp\left(j\phi_{i_r}\right)\exp\left(-j\pi K_r\widetilde{t}_r{}^2\right)\exp\left[j\pi K_r\widetilde{t}_r\left(\alpha_{i_r+1}^{R}+\alpha_{i_r}^{R}\right)\right]\exp\left(j\pi K_r\frac{1}{T_r}\right)\\ &\quad+\frac{1-\beta}{2}\sum_{i_r=1}^{N_r}\Delta\alpha_{i_r}^{R}\operatorname{sinc}\left[\left(K_r\widetilde{t}_r-\frac{1}{\Delta\alpha_{i_r}^{R}}\right)\Delta\alpha_{i_r}^{R}\right]\exp\left(j\phi_{i_r}\right)\exp\left(-j\pi K_r\widetilde{t}_r{}^2\right)\exp\left[j\pi K_r\widetilde{t}_r\left(\alpha_{i_r+1}^{R}+\alpha_{i_r}^{R}\right)\right]\exp\left(-j\pi K_r\frac{1}{T_r}\right)\end{aligned} \tag{48}$$

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TABLE I
MAIN SYSTEM PARAMETERS OF THE SIMULATION

Parmeters	Values
Carrier frequency	16.8 GHz
Flight speed	90 m/s
Reference slant range	4.05 Km
System actual PRF	180 Hz
Bandwidth	300 MHz
Pulse duration time	2.89 us
Number of channels	3
Channel spacing	0.6 m

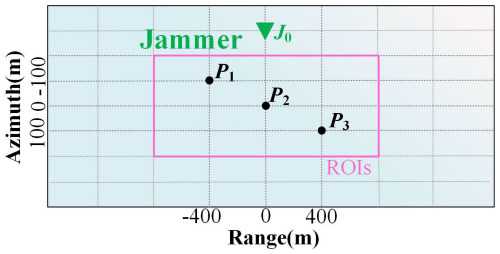


Fig. 7. Point scatterer model.

C. System Parameter Error Analysis

The PCHS-SP method proposed in this article is coherent jamming, and its implementation depends on the accuracy of parameter reconnaissance. Therefore, analyzing the jamming effect under reconnaissance errors of key parameters is essential to guide the formulation of the next operational plan.

1) Position Measurement Error: Assuming the preset jamming position and the actual position have errors in azimuth and ground range directions, denoted as ε_r and ε_a, respectively. The range frequency modulation amount and azimuth delay amount can be expressed as

$$\begin{cases} \Delta\mathbf{P_r} = \exp\left[j2\pi K_r t_r\left(\dfrac{y_p - y_i + \varepsilon_r}{c}\right)\right] & \text{Range} \\ \Delta\mathbf{P_a} = \lceil\left(\left|\dfrac{x_p - x_i + \varepsilon_a}{v_a}\right| - \mu\right)\cdot\mathbf{F}_p\rceil & \text{Azimuth.} \end{cases} \tag{61}$$

Therefore, if there is a position reconnaissance error, the preset jamming position will deviate from the actual position, and the magnitude of the deviation is directly proportional to the error amount.

2) Speed Error: If there is a velocity error V_e, then (24) can be further expressed as

$$R_m^J(\eta, V_e) = 2R_0 - \frac{(v_a + V_e)x_j + D_m}{R_0}\eta + \frac{(v_a + V_e)^2}{R_0}\eta^2 + \frac{2x_j D_m + D_m^2 + x_j^2}{2R_0}. \tag{62}$$

From (62), the speed errors exist in the linear term and the quadratic term of V_e, respectively. The existence of linear term error will cause the azimuth jamming position to deviate. The measurement error of the quadratic term will further cause the azimuth modulation frequency mismatch. For barrage jamming, the influence of the quadratic term of V_e can be compensated by appropriately changing the expansion factor. However, for deception jamming, the false target will inevitably produce defocus. Therefore, we need to specify jamming decisions based on the appropriate ECM environment.

3) Carrier Frequency Error: Finally, the carrier frequency error is considered to acquire better jamming performance. Usually, the carrier frequency error has no impact on the range direction. From (31), assuming there is a carrier frequency error f_e, it should appear in the linear term, quadratic term, and constant term, respectively. The constant term is usually ignored. Therefore, similar to the effects of velocity errors, carrier frequency errors will also lead to azimuth position deviation and mainlobe defocusing. Without loss of generality, only the effects of SAR location error, velocity error, and carrier frequency error are detailed in this section, which will be taken into account in the implementation of the PCHS-SP method.

V. SIMULATION RESULTS AND ANALYSIS

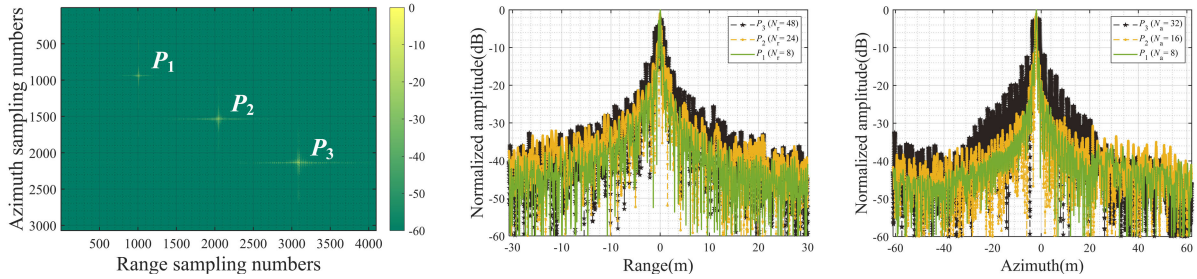
To verify the effectiveness of the method proposed in this article, this section conducts simulation experiments on point targets and area targets, respectively. Meanwhile, the jamming effect under parameter error is considered. In addition, comparison with other methods fully proves the superiority of the proposed PCHS-SP method. The main system parameters used in the simulation are listed in Table I.

A. Barrage Jamming Parameters on Performance Analysis

To verify the effectiveness of the proposed PCHS-SP method in barrage jamming, a simulation scene with a three-point scatterer array is set, as shown in Fig. 7. The jammer is arranged at a distance from the ROIs, that is, $J_0(-300m, 0)$. Meanwhile, a three-point scatterer array is

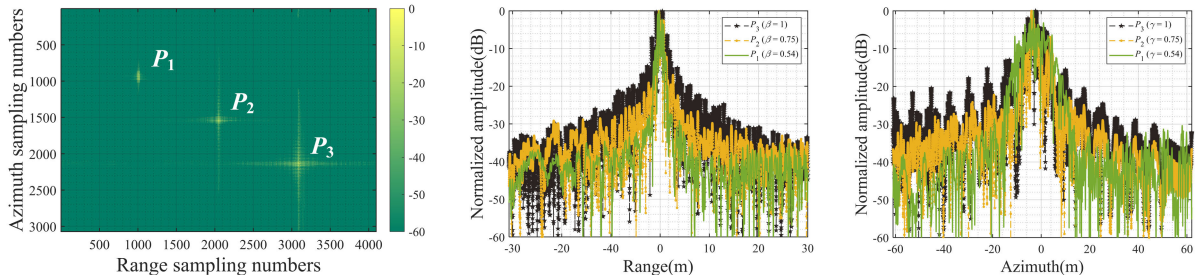
$$\begin{aligned} &\widetilde{S}_{i_a}^{o}(\widehat{\eta}) \\ &= \gamma\Delta\widehat{\alpha}_{i_a}^{A}\mathrm{sinc}\left(K_a\widetilde{\eta}\Delta\widetilde{\alpha}_{i_a}^{A}\right)\exp\left(j\phi_{i_a}\right)\exp\left(-j\pi K_a\widetilde{\eta}^{\,2}\right)\exp\left[j\pi K_a\widetilde{\eta}\left(\widehat{\alpha}_{i_a+1}^{A} + \widehat{\alpha}_{i_a}^{A}\right)\right] \\ &\quad + \frac{1-\gamma}{2}\Delta\widehat{\alpha}_{i_a}^{A}\mathrm{sinc}\left[\left(K_a\widetilde{\eta} + \frac{1}{\Delta\widehat{\alpha}_{i_a}^{A}}\right)\Delta\widehat{\alpha}_{i_a}^{A}\right]\exp\left(j\phi_{i_a}\right)\exp\left(-j\pi K_a\widetilde{\eta}^{\,2}\right)\exp\left[j\pi K_a\widetilde{\eta}\left(\widehat{\alpha}_{i_a+1}^{A} + \widehat{\alpha}_{i_a}^{A}\right)\right]\left(j\pi\frac{\widehat{\alpha}_{i_a+1}^{A} + \widehat{\alpha}_{i_a}^{A}}{\Delta\widehat{\alpha}_{i_a}^{A}}\right) \\ &\quad + \frac{1-\gamma}{2}\Delta\widehat{\alpha}_{i_a}^{A}\mathrm{sinc}\left[\left(K_a\widetilde{\eta} - \frac{1}{\Delta\widehat{\alpha}_{i_a}^{A}}\right)\Delta\widehat{\alpha}_{i_a}^{A}\right]\exp\left(j\phi_{i_a}\right)\exp\left(-j\pi K_a\widetilde{\eta}^{\,2}\right)\exp\left[j\pi K_a\widetilde{\eta}\left(\widehat{\alpha}_{i_a+1}^{A} + \widehat{\alpha}_{i_a}^{A}\right)\right]\left(-j\pi\frac{\widehat{\alpha}_{i_a+1}^{A} + \widehat{\alpha}_{i_a}^{A}}{\Delta\widehat{\alpha}_{i_a}^{A}}\right) \end{aligned} \tag{54}$$

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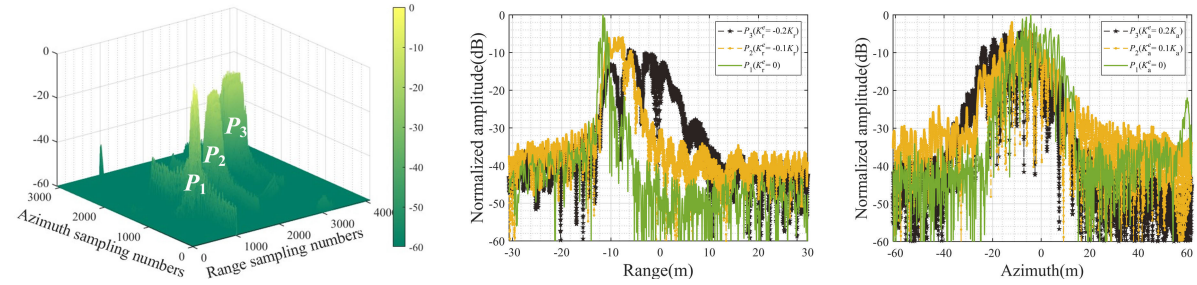
(a) (b) (c)

Fig. 8. Simulation results of $P_1(N_r = 8, N_a = 8)$, $P_2(N_r = 24, N_a = 16)$, and $P_3(N_r = 48, N_a = 32)$ three-point scatterer with different number of segments in 2-D direction (without piecewise weighting). (a) Imaging results of three-point scatterers. (b) Range slices of P_1, P_2, and P_3 points. (c) Azimuth slices of P_1, P_2, and P_3 points.



(a) (b) (c)

Fig. 9. Simulation results of $P_1(\beta = 0.54, \gamma = 0.54)$, $P_2(\beta = 0.75, \gamma = 0.75)$, and $P_3(\beta = 1, \gamma = 1)$ three-point scatterer with different weighting coefficient in 2-D direction ($N_r = N_a = 24$). (a) Imaging results of three-point scatterers. (b) Range slices of P_1, P_2, and P_3 points. (c) Azimuth slices of P_1, P_2, and P_3 points.



(a) (b) (c)

Fig. 10. Simulation results of $P_1(K_r^e = 0, K_a^e = 0)$, $P_2(K_r^e = -0.1K_r, K_a^e = 0.1K_a)$, and $P_3(K_r^e = -0.2K_r, K_a^e = 0.2K_a)$ three-point scatterer with different expansion factor in 2-D direction ($N_r = N_a = 24$, $\beta = \gamma = 0.54$). (a) Imaging results of three-point scatterers. (b) Range slices of P_1, P_2, and P_3 points. (c) Azimuth slices of P_1, P_2, and P_3 points.

set within the ROIs area, with the range direction distance between each point scatterer being 400 m and the azimuth direction distance being 100 m. In addition, the subsegment divisions in the range and azimuth directions are equally divided.

1) Number of Subsegments: To verify the impact of the number of subsegments against barrage jamming, Fig. 8 shows the imaging results of the three-point scatterer P_1, P_2, and P_3 after random phase modulation with different subsegment numbers. Fig. 8(a) shows the 2-D imaging results of a three-point scatterer, which generates a jamming effect only at specified locations. This preliminarily verifies the effectiveness of the proposed PCSH-SP method against the HRWS-SAR, which effectively avoids the occurrence of azimuth periodic repetitions. Furthermore, Fig. 8(b) and Fig. (c) shows the range and azimuth slices of points P_1, P_2, and P_3, respectively. Obviously, as the number of subsegments increases, the mainlobe becomes wider, that is, the jamming range increases. Especially for a point P_3, the jamming ranges of the corresponding 2-D point spread function in the range and azimuth direction are 2.50 and 2.84 m, respectively. This implies that control over the jamming range can be achieved within a smaller area by controlling the number of subsegments. However, increasing the number of segments leads to a significant rise in the sidelobe, resulting in the dispersion of jamming energy. Taking Fig. 8(b) as an example, the peak sidelobe ratios (PSLR) of P_1, P_2, and P_3 are -17.3, -15.8, and -14.4 dB, respectively. Hence, the influence of the number of subsegments for barrage jamming is verified by simulation. The increased number of subsegments will make the 2-D mainlobe of the point spread function broaden and the sidelobe rise.

2) Weighting Coefficient: From Section IV-A2, piecewise weighting can reduce the sidelobes so that the jamming

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TABLE II
PSRL UNDER DIFFERENT WEIGHTING COEFFICIENTS CORRESPONDING TO P_1, P_2, AND P_3 THREE-POINT SCATTERERS

Weighting coefficient	PSLR (dB)	
	Range	Azimuth
$P_3(\beta = \gamma = 1)$	−10.6	−13.2
$P_2(\beta = \gamma = 0.75)$	−20.3	−22.1
$P_1(\beta = \gamma = 0.54)$	−33.1	−35.2

TABLE III
SIMULATION AND THEORETICAL ANALYSIS RESULTS COMPARISON

False target	Position(m)		Amplitude(dB)	
	Theoretical	Simulation	Ttheoretical	Simulation
P_4	(0, 0)	(−0.3, 0)	0	0
P_5	(−37, 37)	(−37.9, 36.6)	−12	−11.8
P_6	(35, −35)	(36.2, −35.3)	−12	−11.6

energy is concentrated on the mainlobe. Taking the number of range and azimuth subsegments $N_r = N_a = 24$ as an example, Fig. 9 shows the simulation results under different weighting coefficients. Fig. 9(a) shows the imaging results of the three-point scatterer P_1, P_2, and P_3. It can be seen that piecewise weighting in the azimuth frequency domain also avoids the occurrence of azimuth periodic repetitions. Fig. 9(b) and (c) shows the corresponding range and azimuth slices, respectively. The 2-D mainlobe of the point spread function has been suppressed to varying degrees. The correctness of the theoretical analysis in Section IV-A2 is verified. Furthermore, Table II gives the quantitative evaluation results. If $\beta = \gamma = 1$, that is, without piecewise weighting, the sidelobes of the point spread function have been raised to about −10 dB. As the weighting coefficient decreases, the sidelobes are suppressed to varying degrees. Particularly, if $\beta = \gamma = 0.54$, that is, hamming window, the mainlobe of the point spread function is suppressed to about −34 dB. The influence of sidelobes is basically eliminated. The effectiveness of piecewise weighting is verified.

3) Expansion Factor: Although increasing the number of subsegments can expand the jamming range to some extent, it cannot achieve precise 2-D control of the jamming range within a large dynamic range. Taking the number of segments $N_r = N_a = 24$ and the weighting coefficient $\beta = \gamma = 0.54$ as an example, Fig. 10 shows the simulation results under different chirp rate errors. Fig. 10(a) shows the 3-D imaging result. Obviously, the increase in chirp rate error significantly increases the jamming range. Fig. 10(b) and (c) shows the range and azimuth slices corresponding to the three points, respectively. After calculation, the jamming ranges corresponding to P_1, P_2, and P_3 are 2.6, 7.1, and 14.6 m in the range direction, respectively. The jamming ranges corresponding to P_1, P_2, and P_3 are 37.4, 41.2, and 45.8 m in the azimuth direction, respectively. This is consistent with the theoretical value calculated by (29) and (35). Therefore, 2-D precisely controllable barrage jamming within a large dynamic range can be achieved by controlling the expansion factor. However, increasing the jamming range will reduce the gain of PC processing of jamming signals. Therefore, the selection of the expansion factor needs to be based on the actual ECM combat environment.

Through the above simulation experiments, the proposed PCSH-SP method can produce noise-like jamming effects in the 2-D mainlobe range of the point spread function and has low sidelobe characteristics. Meanwhile, a 2-D accurately controllable barrage jamming range can also be realized by adjusting the expansion factor.

B. Deceptive Jamming Parameters on Performance Analysis

To verify the effectiveness of deceptive jamming, this section performs the periodic phase modulation proposed in this article on the scatterer at the position of point P_2. Fig. 11(a) shows the result of modulation of a single scattering point in the range direction (set parameters as $\theta_{l_r}^R = [\exp(j0.2\pi), \exp(-j0.3\pi)]$, $N_r = 2048$). According to Fig. 11(a), a series of false target strings are generated in the range direction, and the amplitude of the false target strings fluctuates greatly. A realistic deceptive jamming effect is produced in the range direction. Furthermore, the theoretical derivation results of the false target number are consistent. Fig. 11(b) shows the result of modulation of a single scattering point in the azimuth direction (set parameters as $\theta_{l_a}^A = [\exp(j0.2\pi), \exp(-j0.3\pi)]$ and $N_a = 1024$). Similar to the range direction, a series of false targets are generated in the azimuth direction. More importantly, the azimuth false target is not subject to the azimuth periodic repetition and sin unction weighting caused by the multichannel reconstruction. The effectiveness of the proposed PCSH-SP method for deceptive jamming has been preliminarily verified. Based on the parameter settings in Fig. 11(a) and (b), Fig. 11(c) shows the imaging effect if the jamming echo is modulated simultaneously in the range and azimuth directions. Obviously, multiple false targets were generated simultaneously in the range and azimuth directions, similar to a false target group array. However, the amplitude of edge false targets is lower, which is consistent with the analysis result of (56). Subsequently, Fig. 11(d) shows the simulation results after random periodic phase modulation, 2-D false target groups with random fluctuations in amplitude are generated, and there are no rules to follow, that is, a highly concealed deceptive effect is formed. Therefore, the amplitude of the false target can be adjusted to a certain extent by changing the phase value of the modulation.

To verify the influence of subsegment number and modulation period, Fig. 11(e) and (f) shows the simulation results under different parameters, respectively (RMP and RAP represent the range and azimuth modulation periods, respectively). First, comparing the range directions in Fig. 11(e) and (f), we can see that the increase of the modulation period will increase the number of false targets. When comparing the azimuth directions in Fig. 11(e) and (f), it is evident that increasing the modulation period also results in a higher number of false targets. Subsequently, to further

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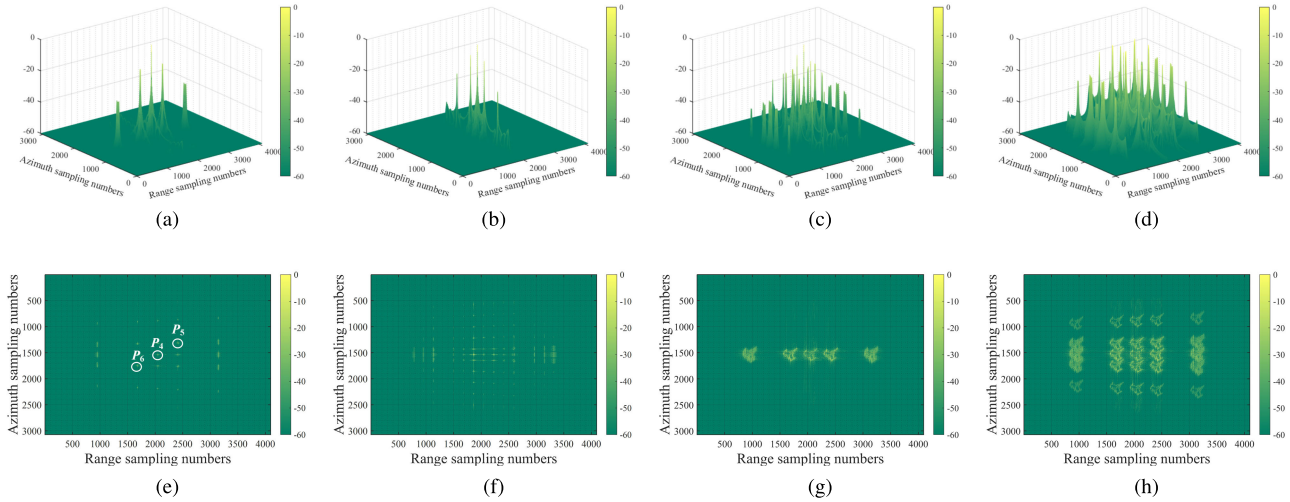


Fig. 11. Simulation results of deceptive jamming. (a) 3-D imaging results with range direction modulation ($L_r = 2048$). (b) 3-D imaging results with azimuth direction modulation ($L_a = 1024$). (c) 3-D imaging results of periodic phase modulation in the 2-D direction ($L_r = 2048$ and $L_a = 1024$). (d) 3-D imaging results of random periodic phase modulation in the 2-D direction. (e) 3-D imaging results of random periodic phase modulation in the 2-D direction (RMP $= 2\tau_r$, AMP $= 2\tau_a$, $L_r = 2048$, and $L_a = 1024$). (f) 2-D imaging results of random periodic phase modulation in the 2-D direction (RMP $= 4\tau_r$, AMP $= 2\tau_a$, $L_r = 2048$, and $L_a = 512$). (g) 2-D imaging results of multipoint scatterers with range direction periodic phase modulation. (h) 2-D imaging results of multipoint scatterers with 2-D periodic phase modulation.

evaluate the effectiveness of the proposed method, three false target points P_4, P_5, and P_6 in Fig. 11(e) were selected for theorem analysis. Table III shows the simulation results and theoretical derivation results. The theoretical values and simulation values are basically consistent, which proves the correctness of the theoretical analysis. To further verify the effectiveness of the proposed PCSH-SP deceptive jamming, we also use a plane model with multiple scattering points for simulation, as shown in Fig. 11(g) and (h). The generated 2-D false target image is almost a replica of the real target image, but the imaging quality of edge false targets is poor.

From the above simulation experiments, the proposed PCHS-SP can generate 2-D accurately controllable false targets. Meanwhile, the false target has highly adjustable features by controlling the modulation period, number of segments, and phase value, which provides a basis for target protection in complex and changeable ECM.

C. Jamming Strategy With Parameter Estimation Error

Generally, the parameters provided by the electronic warfare system will unavoidably deviate from the actual ones. Therefore, the proposed PCHS-SP method needs to consider the jamming performance in the presence of parameter errors to formulate an appropriate ECM strategy.

Assume that the preset jamming position deviates from the jammer by 50 m in both the range and azimuth directions, the jammer performs delayed forwarding and frequency shift modulation against the intercepted SAR signal to achieve the desired ideal effect at the desired position. For the position measurement error, since the preset jamming position is usually known in advance, the error is generally smaller. Based on this, Fig. 12(a) shows the distance error between the ideal and actual jamming positions when the distance and azimuth measurement errors are between 0% and 10%. The offset distance gradually increases as the position measurement error increases. However, even when the measurement error reaches 10%, the range and azimuth offset distances only deviate by about 1.5 m. Usually, it appears as a deviation of several pixels on the SAR image. Therefore, position measurement error has less impact on the jamming position. From Section IV-C, the carrier frequency and speed errors affect the azimuth jamming position and chirp rate, respectively. Therefore, Fig. 12(b) and (c), respectively, shows the impact of speed error and carrier frequency error between $[-20\%, 20\%]$ on the jamming location and jamming range. In fact, with the continuous development of ECM systems, the reconnaissance accuracy of electronic reconnaissance systems can usually reach within 10%. The parameter error range selected in this article is considered from the boundary conditions. From Fig. 12(b), the speed error and the carrier frequency error cause the jamming position to deviate, but within a smaller range and usually can be ignored. From Fig. 12(c), the carrier frequency error and speed error will cause the azimuth jamming range to further increase. For barrage jamming, the expansion of the jamming range caused by parameter errors can be compensated by adjusting the value of the expansion factor, thereby achieving precise controllability of the 2-D jamming range. For deceptive jamming, parameter error causes the false target to defocus. Therefore, the deceptive jamming effect of the proposed PCSH-SP method has certain limitations, which depend on the accuracy of parameter reconnaissance.

In summary, parameter errors will cause deviations in the jamming position and jamming range in the azimuth direction. The theoretical analysis has been verified and provided certain theoretical guidance for formulating appropriate combat plans in complex ECM environments.

D. Area Simulation and Comparison of Different Jamming Methods

To further verify the effectiveness of the proposed method against HRWS-SAR systems, area target simulation experiments are further introduced in this section. Meanwhile,

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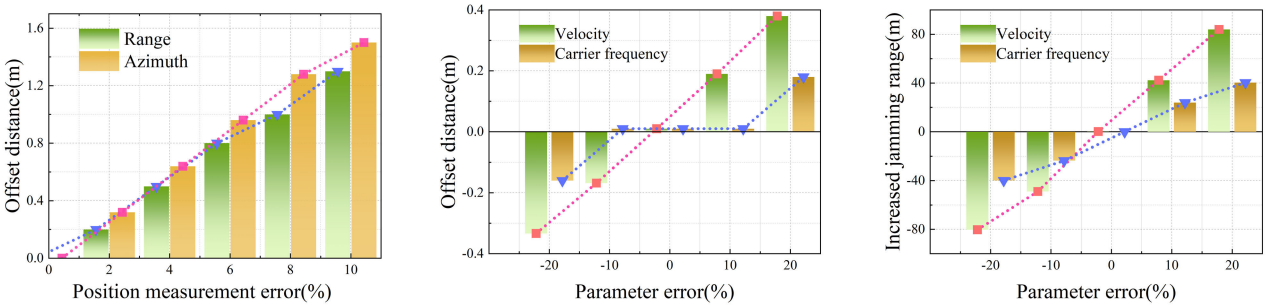


Fig. 12. Verification of the theoretical error model using the point-like target simulations. (a) Relationship between position measurement error and offset (between the preset jamming position and the actual jamming position). (b) Relationship between parameter error and offset. (c) Relationship between parameter error and increased jamming range.

to verify the superiority of the proposed method against the HRWS-SAR system, some advanced jamming methods, including noise convolution modulation jamming (NCMJ) [24], parameterized and large-dynamic-range 2-D precise controllable (PLDR-PC) jamming [34], PEM jamming [33], and smart jamming based on nonlinear frequency-modulated signal (NLFM-SJ) [32], were selected as comparative experiments.

Fig. 13 shows the simulation results of the area target. An airborne SAR image from Sandia National Laboratories was selected as the real scene, as shown in Fig 13(a). Meanwhile, to further highlight the 2-D precision controllable ability of the proposed PCSH-SP method, three protection areas ①, ②, and ③ with different jamming positions and jamming ranges in the ROIs were selected as jamming targets. Fig. 13(b) shows the barrage jamming effect of the PCSH-SP method on area ① ($N_a = 24, N_r = 64, \beta = \gamma = 1, K_r^e = 0.8K_r$, and $K_a^e = 0.95K_a$). Obviously, the noise-like jamming effect is generated in the target area ①. However, due to the influence of high sidelobes, the noise resembles jamming and diffuses outside the target area ①, which results in a waste of jamming energy. From Section V-A, if the weighting coefficient is 0.54, that is, the hamming window, the sidelobe suppression effect is the best. Therefore, based on Fig. 13(b) jamming parameter, Fig. 13(c) shows the simulation results after piecewise weighting (weighting coefficient is 0.54). Obviously, the energy diffusion caused by high-sidelobes is eliminated, and the jamming energy is concentrated within the jamming range. The jamming effect of piecewise weighting has been further verified. Fig. 13(d) and (e) shows the barrage jamming effect on areas ② and ③, respectively. The vehicle military targets in target areas ② and ③ have been basically submerged in noise. Meanwhile, the noise-like jamming piece generated is the same as the size of area ② and ③, which means that the PCSH-SP method can achieve 2-D precise control of the jamming range and jamming location in the ROIs area. In fact, due to the independence of range and azimuth, the proposed method can simultaneously achieve barrage and deceptive jamming effects. As shown in Fig. 13(f), a series of equidistant noise lines are generated in the range direction, that is, range deceptive jamming and azimuth barrage jamming. In particular, a controllable barrage jamming with a large dynamic range can also be achieved by controlling the expand factor. Fig. 13(g) shows the simulation results of parameters under $N_a = 64, N_r = 256, \beta = \gamma = 0.54, K_r^e = 0.8K_r$, and $K_a^e = 0.5K_a$, which noise-like jamming effects are produced in the entire ROIs area. However, the increase in the jamming range will reduce the PC gain of the jamming signal.

Taking ③ in the ROIs area as an example, Fig. 13(h)–(j) shows the simulation results of NCMJ, PLDR-PC, and PEM, respectively. From Fig. 13(h), although the NCMJ method can achieve an effective barrage of ③, it is affected by multichannel reconstruction, in which the noise template produces periodic repetition sums and weighting of sinusoidal functions in the azimuth, as shown in the white box in Fig. 13(h). Similarly, due to the influence of azimuth multichannel reconstruction, PLDR-PC and PEM also produce periodic repetitions in the azimuth direction, as shown in Fig. 13(i) and Fig. (j). The unique azimuth multichannel periodic repetition not only wastes jamming energy but also increases the possibility of exposing the jammer due to the regular appearance of interference blocks. Fig. 13(k) is the simulation result of NLFM-SJ. Although NLFM-SJ effectively barrage area ③ and is not affected by multichannel reconstruction, it is only discussed in terms of the range direction. Moreover, NLFM-SJ cannot achieve precise 2-D control of the jamming location.

All in all, the proposed method can effectively avoid the azimuth periodic repetition and the weighting of sinusoidal functions caused by multichannel reconstruction processing, which generates 2-D accurately controllable smart jamming in the ROIs area. Furthermore, PCSH-SP provides a feasible option to countermeasure the HRWS-SAR system.

E. Algorithm Complexity Analysis

Finally, this section discusses the computational complexity of the proposed jamming technique. From Section III, the computational complexity of the PCHS-SP mainly includes multiplication and fast Fourier transform. Meanwhile, it is assumed that the number of sampling points in the range and azimuth directions of the intercepted SAR signal is N_r^E and N_a^E, respectively. For barrage jamming, the first modulation step is random phase modulation. This is mainly manifested as a 2-D point-to-point product, so its corresponding computational complexity is $O(N_r^E \cdot N_a^E)$. The next step is to perform

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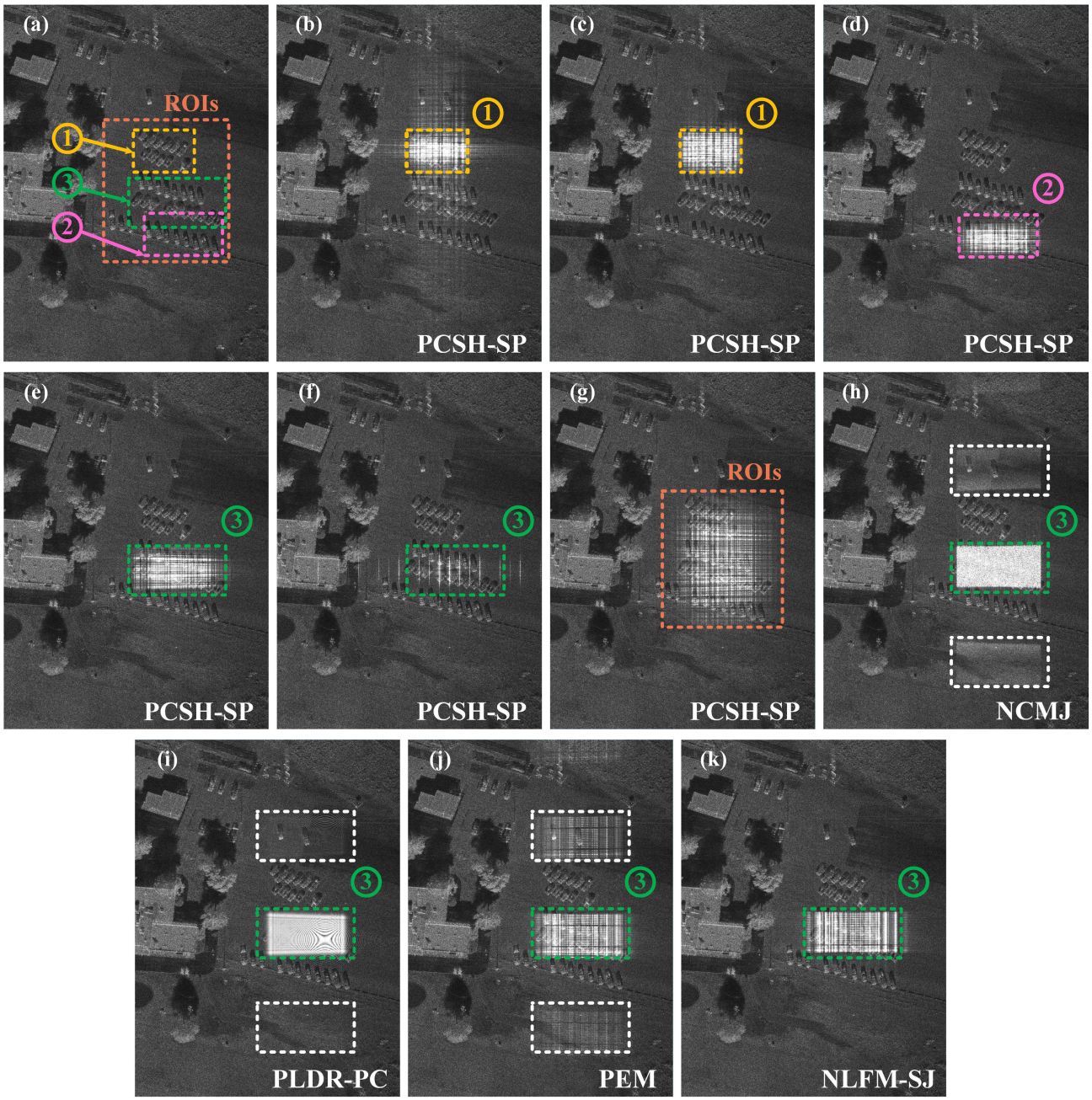


Fig. 13. Area target simulation results. (a) Real scene without jamming. (b) Simulation results of the PCSH-SP method in area ① ($N_a = 24, N_r = 64, \beta = \gamma = 1, K_r^e = 0.8K_r$, and $K_a^e = 0.95K_a$). (c) Simulation results of the PCSH-SP method in area ① ($N_a = 24, N_r = 64, \beta = \gamma = 0.54, K_r^e = 0.8K_r$, and $K_a^e = 0.95K_a$). (d) Simulation results of the PCSH-SP method in area ② ($N_a = 16, N_r = 160, \beta = \gamma = 0.54, K_r^e = 0.8K_r$, and $K_a^e = 0$). (e) Simulation results of the PCSH-SP method in area ③ ($N_a = 16, N_r = 256, \beta = \gamma = 0.54, K_r^e = 0.8K_r$, and $K_a^e = 0.85K_a$). (f) Simulation results of the PCSH-SP method in area ③ ($N_a = 16, N_r = 512, \beta = \gamma = 0.54, K_r^e = 0$, and $K_a^e = 0$). (g) Simulation results of the PCSH-SP method in ROIs area ($N_a = 64, N_r = 256, \beta = \gamma = 0.54, K_r^e = 0.8K_r$, and $K_a^e = 0.5K_a$). (h) Simulation results of NCMJ. (i) Simulation results of PLDR-PC. (j) Simulation results of PEM. (k) Simulation results of NLFM-SJ.

segment weighting in the RD domain. Since this part involves the Fourier transform and inverse Fourier transform, the corresponding computational complexity is $O(2 \cdot N_a^E log_2(N_a^E) + N_a^E \cdot N_r^E)$. Finally, the computational complexity of barrage jamming can be summarized as

$$O\left(N_a^E log_2\left(N_a^E\right)\right) + O\left(N_a^E \cdot N_r^E\right). \tag{63}$$

For deceptive jamming, since it does not involve fast Fourier transform and inverse Fsat transform, the corresponding calculation complexity is

$$O\left(N_a^E \cdot N_r^E\right). \tag{64}$$

VI. Conclusion

In this article, a 2-D precise and controllable smart jamming technique, that is, PCSH-SP, is proposed against the unique multichannel reconstruction mechanism of the HRWS-SAR

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system. The PCSH-SP avoids the effects of azimuthal periodicity and sin functions caused by multichannel reconstruction processing through the concept of presegmentation, enabling jamming to be more effectively concentrated within the target range.

The PCSH-SP can produce two jamming effects: barrage and deceptive jamming. For barrage jamming, a variety of barrage effects can be achieved by adjusting parameters such as the number of segments, weighting coefficients, and expansion factors. Additionally, a precise and controllable jamming range can be achieved by establishing the relationship between chirp rate error and jamming range. For deceptive jamming, flexible jamming effects can be achieved by adjusting parameters including the number of segments, phase values, and modulation period. Finally, the superiority of the proposed method in countering HRWS-SAR systems has been verified through comparison with other methods. For the jamming against the opponent's SARs, future studies might focus on the following aspects: 1) exploring precise control of the azimuthal jamming position from the signal domain and 2) studying dynamic target jamming technology in the context of the HRWS-SAR system.

APPENDIX A

Assuming that the number of sampling points in each subsegment is n_p, the segmentation modulation jamming term mentioned in this article can be expressed as

$$\mathbf{S_1}(\eta) = \mathbf{X}(\eta) \cdot \mathbf{Y} \tag{65}$$

where $\mathbf{X}(\eta)$ can be expressed as

$$\mathbf{X}(\eta) = \begin{cases} 1 & -\dfrac{T_a}{2 \cdot M} < \eta < \dfrac{T_a}{2 \cdot M} \\ 0 & \text{otherwise.} \end{cases} \tag{66}$$

$\mathbf{Y}$can be expressed as

$$\mathbf{Y} = \sum_{i=1}^{N_a} \left[\epsilon(\eta - a_i) - \epsilon(\eta - a_{i+1})\right] \exp(j2\pi\phi_i). \tag{67}$$

The Fourier transform of $\mathbf{S_1}(\eta)$ can be expressed as

$$\mathbf{S_2}(f_a) = \int_{-\infty}^{\infty} \mathbf{S_1} e^{-j2\pi\eta} d\eta = \int_{-\infty}^{\infty} \mathbf{X}(\eta) \cdot \mathbf{Y} e^{-j2\pi\eta} d\eta. \tag{68}$$

From the above equation, we can observe that $\mathbf{Y}$ is a composite signal composed of multiple unique constant phases in different subsegments. Based on this observation, we can simplify the above formula and express it as the accumulation or combination of the constant phase contributions in each subsegment. Therefore, (68) can be simplified to

$$\mathbf{S_3}(f_a) = \sum_{i=1}^{N_a} \left[\epsilon(\eta - a_i) - \epsilon(\eta - a_{i+1})\right] \exp(j2\pi\phi_i) \frac{T_a}{M} \text{sinc}(f_a T_a / M). \tag{69}$$

From the above equation, in the framework of in-depth analysis of single-channel data processing, the segment modulation function introduced in this manuscript exhibits a unique characteristic in spectral representation: its spectral morphology basically follows the outline of the "sinc" function. However, in each subsegment, this basic spectrum is modulated by a specific constant phase. The effectiveness of the proposed method has been preliminary verified. Next, consider the spectrum of the segment-modulated jamming function after multichannel reconstruction, which can be expressed as

$$\mathbf{S_1^e}(\eta) = \mathbf{X^e}(\eta) \cdot \mathbf{Y^e} \tag{70}$$

where $\mathbf{X^e}(\eta)$ can be expressed as

$$\mathbf{X^e}(\eta) = \begin{cases} 1 & -\dfrac{T_a}{2} < \eta < \dfrac{T_a}{2} \\ 0 & \text{otherwise.} \end{cases} \tag{71}$$

$\mathbf{Y^e}$can be expressed as

$$\mathbf{Y^e} = \sum_{i=1}^{N_a} \left[\epsilon(\eta - M \cdot a_i) - \epsilon(\eta - M \cdot a_{i+1})\right] \exp(j2\pi\phi_i). \tag{72}$$

After Fourier transformation, $\mathbf{S_1^e}(\eta)$ can be expressed as

$$\mathbf{S_3^e}(f_a) = \sum_{i=1}^{N_a} \left[\epsilon(\eta - a_i) - \epsilon(\eta - a_{i+1})\right] \exp(j2\pi\phi_i) T_a \text{sinc}(f_a T_a) \tag{73}$$

The above equation shows the frequency-domain expression of the segment modulation function after multichannel reconstruction. Obviously, the spectrum expression of the segmented modulation function after multichannel reconstruction is basically the same as before, and both can be regarded as a "sinc" function modulated by a different constant phase. However, compared with before multichannel reconstruction, the spectrum amplitude of the segmented modulation function after multipass reconstruction is expanded by M times. This confirms that the segment modulation function proposed in this manuscript can be regarded as a "sinc" function modulated by a different constant phase before and after multichannel reconstruction, that is, multichannel reconstruction does not affect the spectrum of the proposed segment modulation function. Finally, the above problems are avoided.

ACKNOWLEDGMENT

The authors would like to thank both the anonymous reviewers and the editors for the valuable comments and suggestions that helped improve the clarity of this article.

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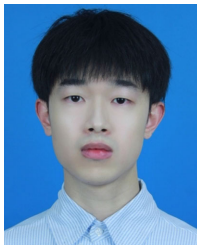


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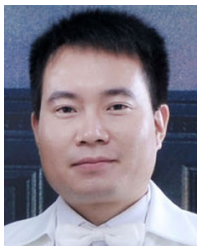
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